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Crazy 1

Papers 1–4 combined · Sections 1–3

All four answer books merged section by section; overlapping answers are printed once with every fact kept, tagged P1–P4 by where they were asked.

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Section 1 · Basic Principles

Long answer questions (LAQ)

LAQ 1 Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy

P1 **merged from 3**

Asked as: P1 LAQ "Iodinated and other contrast agents (use in urology) and their complications" · P1 SAQ "Non iodinated contrast materials" · P1 SAQ "Acetylcysteine and Contrast Induced Nephropathy (CIN)"

Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All iodinated agents are built on a **tri-iodinated benzene ring**.
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.
- Non-iodinated agents are used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

Classification of iodinated agents

Class	Example	Osmolality (mOsm/kg)	Iodine : particles
High-osmolar ionic monomer	Diatrizoate	~1500–2000 (5–8 × plasma)	3 : 2
Low-osmolar non-ionic monomer	Iohexol, iopamidol	~600–850	3 : 1
Low-osmolar ionic dimer	Ioxaglate	~600	6 : 2
Iso-osmolar non-ionic dimer	Iodixanol	~290	6 : 1

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

Types of contrast agent and their uses in urology

Agent	Principle	Urological use
Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg)	Iodine absorbs X-rays; glomerular filtration	CT urography, angiography; retrograde pyelogram, urethrogram, cystogram
Gadolinium (GBCA, gadolinium-based contrast agent)	Shortens T1 relaxation; chelated	MRI of renal mass, prostate, MR urography
Microbubbles	Gas in shell, intravascular	CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography

Agent	Principle	Urological use
Carbon dioxide	Negative contrast	Angiography in CKD (chronic kidney disease)

Complications of iodinated contrast

A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

Type	Timing	Features
Immediate	Within about 1 hour (most within minutes)	Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe
Non-immediate	Hours to days	Maculopapular rash; T-cell mediated, milder

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.
- Retrograde studies are safe in allergy.

B. Contrast-associated acute kidney injury (see the separate CIN section below)

C. Others

- **Metformin:** lactic acidosis only if AKI (acute kidney injury) develops. ACR (American College of Radiology): eGFR (estimated glomerular filtration rate) ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function (creatinine) is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism/thyrotoxicosis (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.

Prevention and treatment of contrast reactions

Premedication for high-risk patients / prior reaction (ACR 2023)

Regimen	Drugs
1	Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before
2	Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before

- First consider ultrasound, MRI or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because little contrast is absorbed.

Treating reactions `Stop injection → airway, oxygen, call for help Urticaria → observe ± H1-antihistamine (may progress) Bronchospasm → salbutamol inhaler 2`

puffs (90 µg each), up to 3 times Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min (repeat), max 1 mg; IV fluids Vasovagal → legs up, IV fluids, IV atropine Observe ≥30 min; later allergy testing; record the reaction

Contrast-induced nephropathy (CIN) and contrast-associated AKI

Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR):
- **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
- **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.

Pathophysiology

- **Renal (afferent) vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury (ischaemic ATN, acute tubular necrosis), plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

Risk factors

- **Patient:** CKD (strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high contrast volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- ACR/NKF (National Kidney Foundation) 2020: older uncontrolled studies overstated the risk; it is rare but real.

Prevention

- Consider ultrasound or non-contrast CT instead.
- Lowest dose of a **low- or iso-osmolar** agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis (removing contrast by dialysis) does not reduce risk.
- **Metformin:** withhold for 48 hours only if eGFR <30 or AKI (see above).

Acetylcysteine

Introduction

- **NAC:** acetylated form of the amino acid cysteine; a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser.

Mechanisms of action

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** in theory forms a soluble mixed disulfide with cystine.

Clinical uses

- **1. Paracetamol (acetaminophen) overdose: main indication.**
- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) builds up once glutathione is exhausted, causing hepatic necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later.
- **IV 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.
- **2. Mucolytic:** COPD (chronic obstructive pulmonary disease), cystic fibrosis.

Urological relevance

- **Contrast-induced AKI:** once widely used (early small trials, KDIGO 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (another thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

Adverse effects

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

Non-iodinated contrast agents

1. Gadolinium-based contrast agents (GBCAs)

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water, giving bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.
- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR <30 mL/min/1.73 m², dialysis, AKI.

ACR group	Agents	NSF risk
I (linear)	Gadodiamide, gadopentetate, gadoversetamide	Highest
II	Gadobenate, gadoteridol, gadoterate, gadobutrol	**Little or none, even in ESRD (end-stage renal disease)**
III	Gadoxetate (liver-specific agent)	Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

2. Ultrasound contrast (microbubbles)

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS:** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD. Intravesical use: **voiding urosonography (ceVUS) for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

3. Other agents

- **CO_2 angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.
- **Also know:** air as negative contrast (historical); radiotracers (^{99m}Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

Diagrams to draw

- Tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.
- Chelated gadolinium (macrocyclic vs linear ring).
- Contrast, then vasoconstriction plus tubular toxicity, then medullary hypoxia, then AKI.
- Paracetamol to NAPQI to glutathione conjugation, with NAC replenishing glutathione.

Key points to remember:

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless; AMACING: no prophylaxis at eGFR 30–59.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic; paracetamol antidote, best within 8 h; cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux. CO_2 angiography: never above diaphragm.

LAQ 2 Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy

P1 · P4 **merged from 2**

Asked as: P1 LAQ "Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy" · P4 SAQ "Different types of trocars in laparoscopy."

Access in laparoscopy

A. Transperitoneal: closed (Veress)

- Veress needle: 14-gauge, spring-loaded blunt stylet. It is an entry needle, not a trocar. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

B. Open (Hasson)

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

C. Retroperitoneal

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas, then **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon), then a balloon-tip port to prevent gas leak and emphysema.

D. Extraperitoneal (pelvic): infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

E. Others: hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

Hasson cannula (Hassan's canula)

Assumed to mean the Hasson cannula, used for open laparoscopic access.

Introduction

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.
- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

Indications

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia. - **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

Instrumentation for development of the extraperitoneal space

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**: finger first, then balloon.

Instrument	Description and use
Open (Hasson) access set	Scalpel, S-retractors, artery forceps
Index finger	Blunt first pocket; feels psoas behind, Gerota's fascia in front
Commercial balloon dissector	Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position
Gaur balloon (1992)	Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments

Instrument	Description and use
Optical (visual) obturator trocar with 0° scope	Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space
Blunt-tip balloon trocar	Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia

Technique: retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.
4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

Technique: pelvic extraperitoneal

- Infraumbilical incision; open anterior rectus sheath, separate recti; finger tunnels behind the rectus to the pubis; pierce transversalis fascia; balloon midline, then each side.

Pitfalls

- Balloon in muscle causes a flank hernia. Too anterior causes peritoneal tear or colon injury; too posterior causes psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal CO_2 needs more ventilation.

Trocars

Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

Classification

Basis	Types
Obturator tip	Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson)
Material	Reusable metal; disposable plastic
Valve	Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)**
Retention	Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)**
Special	Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports

Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (blunt dilating) trocars:** spread muscle fibres; have replaced bladed ones. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching

cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites (all ≥5 mm in children).

- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

Safety points

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

Gases used

Gas	Advantages	Disadvantages
CO_2 (carbon dioxide; standard)	Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed	Absorbed, causing hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease)
Nitrous oxide	Less peritoneal pain, fewer acid–base effects	**Supports combustion**: no electrosurgery
Helium	Inert, no hypercarbia; use in lung disease	Poorly soluble, so embolism risk; costly
Argon	Inert	Embolism risk
Air, oxygen	–	Embolism, combustion: abandoned

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.
- CO_2 absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal CO_2 rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

Parameter	Open RRP	LRP	RALP
Vision	Direct, loupes	2D, unstable	**3D, 10× magnified**, stable

Parameter	Open RRP	LRP	RALP
Operating time	Shortest	Longest early	Longer than open; falls with experience
Instruments	Hands, haptic feedback	Rigid, fulcrum effect	**Wristed** (7 degrees of freedom); no haptics
Learning curve	Moderate	**Steepest** (suturing)	Shorter
Blood loss	300–1000 mL	Less (pneumoperitoneum tamponade)	Least; **transfusion 1–2%**
Hospital stay, catheter	Longer; catheter ~2 weeks	Shorter	Usually 1 day; catheter ~1 week
Pain, return to work	More	Less	Less; normal activity 3–4 weeks
Positive margins	Similar	Similar	Similar (~15%; ~9% if pT2)
Continence (1 year)	~93% (expert)	Similar	Similar; possibly earlier
Potency	Surgeon-dependent	Similar	Similar
Bladder neck stricture	Up to 16% (older series)	Lower	**<2%** (mucosa-to-mucosa)
Unique risks	Wound infection, hernia	Conversion, CO_2 effects	Steep Trendelenburg (eye, nerve), robot failure (<0.5%)
Cost	Lowest	Moderate	**Highest**

Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

Diagrams to draw

- Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP.
- Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.
- Flank section with balloon between psoas and Gerota's fascia.
- Trocar with obturator tip types (bladed, bladeless, optical) and valve.

Key points to remember:

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision; main indication previous surgery or adhesions; avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon. Gaur balloon: glove on red rubber catheter. Balloon in muscle causes hernia.
- Trocar = cannula + obturator. Bladeless: fewer epigastric injuries and hernias. Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

LAQ 3 Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy

P1 · P4 **merged from 2**

Asked as: P1 LAQ "Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy" · P4 SAQ "General complications related to laparoscopy."

"d/t" is read as "due to" and "lap" as laparoscopy.

Introduction to pneumoperitoneum

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both.
- **CO_2** is standard: non-combustible, cheap, highly blood-soluble, so embolism is less lethal. Helium or argon in severe lung disease; N_2O (nitrous oxide) supports combustion.

Choice of insufflant

Gas	Advantages	Disadvantages
CO_2 (standard)	Non-combustible; highly blood-soluble, so embolism is less lethal; cheap	Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it
Helium, argon	Inert; no hypercarbia	Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia
N_2O (nitrous oxide)	Less peritoneal irritation	Supports combustion; no electrosurgery
Room air, oxygen	—	Embolism, combustion; abandoned

Effects by pressure (supine, Campbell Table 18.1)

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Heart rate (HR)	↑	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓
Venous return, cardiac output (Summer 2019 source)	—	Variable	Variable	↓ sharply
GFR (glomerular filtration rate), urine output	↔	↓	↓↓	↓↓
Arterial PCO_2	↔	↑	↑	↑
Arterial pH	↔	↔/↓ (another source: ↔)	↓	↓

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

Cardiovascular (haemodynamic) changes

- **Afterload rises** at every pressure: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.
- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava) and venous return falls. In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone and output collapses.
- **Arrhythmias:** hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may herald pneumothorax, hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous stasis: theoretical DVT (deep vein thrombosis) risk.

Respiratory changes

- **Diaphragm splinted:** FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** (steep Trendelenburg) adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long): pneumothorax, pneumomediastinum.

Acid–base changes (CO_2 absorption)

- CO_2 is absorbed through the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis** with rising end-tidal CO_2. Managed by raising minute ventilation and lowering pressure.
- **COPD** patients cannot blow it off: arterial gases if insufflation exceeds 1 hour, continued in recovery; consider helium. **Hypercarbia peaks after extubation**, as stored CO_2 is released.

Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation.
- Keep pressure ≤10 mmHg (another source: ≤10–12 mmHg); avoid fluid overload. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour, rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

Other systems

- **Splanchnic:** hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.

- **Stress and immune response:** cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery, which may explain faster recovery; oncological benefit unproven.
- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic CO_2 irritation.

Effects of position and approach

Position	Heart rate	MAP	Cardiac output	Other effects
Head-up	↑	↓	↓	Output ↓, hypotension
Head-down (Trendelenburg)	↓	↑	↑ (but lungs worse)	Venous return and output ↑; lung volumes ↓, ICP and IOP ↑
Lateral flank, kidney rest	—	—	—	Lung V/Q mismatch; rhabdomyolysis risk with kidney rest

- **Summary:** head-down raises MAP and output but worsens lungs; head-up lowers output.
- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures than transperitoneal; pelvic extraperitoneal insufflation raised arterial CO_2 more.

Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.
- **Practical rule: work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding); never exceed 20 mmHg for long.
- Monitor end-tidal CO_2, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium (or argon) for lung disease (poorly soluble, so embolism risk).

General complications of laparoscopy

Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

Complications by phase

Phase	Complication	Key points
Access (Veress, first trocar)	Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)**	Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy
Insufflation	**Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia	Embolism: see below
Secondary trocars	Epigastric bleeding; crossing swords, rollover	Transilluminate; bladeless trocars; insert under vision
Dissection	Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury	Four electrosurgical mechanisms below

Phase	Complication	Key points
Positioning	Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal)	Padding, kidney rest <1 h, cases <5 h
Exit	Bowel entrapment, late port bleeding	Inspect each site at 5 mmHg
Postoperative	Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT	Close fascia of large ports

Physiological complications: hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt (see above).

Gas embolism

- Gas enters an opened vein (renal vein, IVC or lumbar vein). Sudden collapse; end-tidal CO_2 rises abruptly, saturation falls, then end-tidal CO_2 falls; "mill-wheel" murmur.
- Treatment: stop insufflation, **left lateral, head down**, 100% oxygen, aspirate through a central line.

Electrosurgical (thermal) bowel injury: four mechanisms

1. **Inappropriate (direct) activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.** - Monopolar injury presents **late (up to 18 days)** with low-grade fever and pain; resect with margin.

Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

Complications of retroperitoneoscopy

Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why its complications differ:** small space with no natural cavity; space made by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; CO_2 is absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.
- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

Classification (by phase of surgery)

Phase	Complications
Access and space creation	Wrong plane, balloon rupture, peritoneal tear, muscle bleeding
Insufflation	Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism
Dissection	Vascular, visceral, nerve, thermal injury

Phase	Complications
Exit and recovery	Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele

1. Access and balloon-related

- **Balloon in muscle** (wrong plane, intramuscular) gives a postoperative **flank hernia** or bulge.
- **Too anterior** gives peritoneal breach or colon injury; **too posterior** gives bleeding from quadratus lumborum or psoas.
- **Balloon rupture:** latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear:** CO_2 enters the peritoneum and collapses the small working space. Fix by venting with a Veress needle or an extra port, enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

2. Gas-related

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally):** pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with needle decompression (16-G needle in the second or third intercostal space, midclavicular line), then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (renal vein injury, IVC or lumbar vein): managed as above.

3. Vascular injury

- Commonest complication of minimally invasive urology; major injury <1 in 1000 (P4: 0.11–2% at access) but fatal in 8–17%. Upper-tract work carries about five times the pelvic vascular risk.
- Sites: renal artery and vein, **lumbar and gonadal veins**, adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding:** raise pressure temporarily to 25 mmHg, suction to see the site, compress or tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, add ports, suture or clip, or convert if uncontrolled. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

4. Visceral injury

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal; usually by thermal spread.
- Thermal bowel injury follows the four mechanisms above; it may present up to 18 days later; diagnose by CT with oral contrast.

5. Positioning and nerve injury

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000

units. Prevent: limit kidney rest to <1 hour, cases <5 hours, avoid hypotension.
- **Nerves** (neuropraxia): brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

6. Postoperative

- Bleeding or haematoma; port-site hernia or haematoma; wound infection.
- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT.
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

Retroperitoneal versus transperitoneal

Retroperitoneal more	Retroperitoneal less
Emphysema, pneumothorax, hypercarbia, disorientation	Bowel injury, ileus, adhesions; peritoneum spared

Prevention

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

Laparoscopic donor nephrectomy

Introduction

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

Donor evaluation `Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → GFR by at least two methods, one measured → metabolic, serology, cancer screening → CT angiography for vessels → cardiac review if needed`

- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

Side selection

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

Techniques

Technique	Features
Pure transperitoneal	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds)	Hand port doubles as extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	No bowel mobilisation; direct hilum; less room

Technique	Features
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed instruments; single incision, less pain

Operative steps (left)

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).
4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes. - **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds. - **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

Advantages versus open

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

Disadvantages and complications

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

Long-term donor counselling

- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.
- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.

Diagrams to draw

- Abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus CO_2 absorption to blood.
- Abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.
- Flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions for retroperitoneoscopy.
- Left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

Key points to remember:

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

LAQ 4 Tissue engineering in urology: principles, status and recent advances

P1 · P3 · P4 **merged from 3**

Asked as: P1 LAQ "Principles of tissue engineering and its use in urology" · P3 SAQ "Describe principles of tissue engineering and what its status in managemet of urological diseases is." · P4 LAQ "Tissue engineering in urology, including recent advances"

Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.
- **Why in urology:** bowel used for bladder and diversion causes mucus, stones, metabolic acidosis, infection and malignancy; grafts for long urethral defects are scarce. The aim is **to replace "wrong tissue" with "right tissue"**.

Principles: the triad plus bioreactor and blood supply

``` Cells + Scaffold + Signals (growth factors, mechanical cycling in a bioreactor) -> construct -> implanted -> vascularised and remodelled by the host

Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals) -> conditioned in bioreactor -> implanted -> host vascularises and remodels ```

## Cell sources

| Source | Features, advantages and limitations |
|---|---|
| **Autologous differentiated cells** (urothelial and smooth muscle cells from a bladder biopsy) | Standard; no rejection; expand well in culture; unusable if the bladder is malignant or diseased |
| **Adult stem cells:** bone marrow or adipose mesenchymal, muscle-derived, **urine-derived** | Easy, non-invasive harvest; act mainly by paracrine (growth factor) effects |
| **Embryonic stem cells** | Pluripotent; ethical issues, teratoma risk |
| **iPSCs** (induced pluripotent stem cells) | Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk |

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

## Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate new tissue forms; porous for cell entry, attachment and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

| Type | Examples | Feature |
|---|---|---|
| **Natural polymers** | Collagen, alginate, silk fibroin | Good cell recognition; weak |
| **Acellular tissue matrices** | Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis | Keep native architecture and stored growth factors |
```

Type	Examples	Feature
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer, poly(lactic-co-glycolic) acid)	Controlled shape and degradation; no bioactivity
Newer	Electrospun nanofibres, hydrogels, bio-ink; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin	Mimic extracellular matrix

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded (acellular)** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

Signals

- Growth factors (for example VEGF, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling to train bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

Vascularisation and innervation (the main barrier)

- Oxygen diffuses only about **100–200 µm**, so thick constructs die centrally without vessels.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in (printed) vascular channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

Clinical applications and status in urology

1. Bladder

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.
- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot yet be recreated.

2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687) (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

3. Urethra

- Acellular bladder matrix onlay for hypospadias and short strictures: success in short onlay repairs.
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

4. Other organs

- **Sphincter, detrusor:** autologous muscle-derived cell injection for SUI (stress urinary incontinence), investigational; for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59).
- **Penis:** engineered corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease), animal or experimental stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

Organ	Status
Bladder	**Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48)
Conduit	Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135)
Urethra	Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity
Sphincter, detrusor	Autologous muscle-derived cells for SUI (investigational); biodegradable scaffold slings; underactive detrusor: open-label improvement in voiding efficiency (Chapter 111)
Vagina	Engineered vaginas in MRKH syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59)
Penis	Corporal tissue and tunica grafts: animal or experimental; stem cells for ED investigational
Kidney	Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices: experimental
Testis	Spermatogonial stem cell banking in prepubertal cancer patients

- **Overall status:** still experimental except short urethral onlays.

Recent advances

3D bioprinting

- Layer-by-layer deposition of **"bio-ink"** (cells in hydrogel), with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.

- Advantages: exact patient-specific shape, several cell types placed precisely, built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models already aid planning and teaching; bioprinted urethra and bladder constructs are still at the animal stage.

Organoids

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: model development and disease, drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing and biology (WNT/β-catenin work, Chapter 129).

Whole-organ decellularisation

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

Other

- Robotic bladder auto-transplantation and robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135): a transplant, not engineered, route to "right tissue" that avoids bowel but needs immunosuppression.
- Smart (growth factor-releasing) scaffolds; urine-derived stem cells; gene-edited and gene-modified cells; cell sheet engineering.

Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval.
- Stem cell tumour risk. Promising pilots followed by disappointing controlled trials.

Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

Diagram to draw

- Biopsy, then cell culture, then seeding on scaffold, then bioreactor, then implant with omental wrap.

> **Key points to remember:**
>
> - Triad: cells, scaffold, signals; plus bioreactor and blood supply.
> - Diffusion limit about 100–200 µm: vascularity is the main barrier.
> - Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
> - Neo-urinary conduit: half stenosed; no blood supply.
> - Raya-Rivera 2011: seeded tubular urethras in 5 boys.
> - Still experimental except short urethral onlays.
> - New: iPSCs, urine-derived cells, bioprinting, organoids.

LAQ 5 Radionuclide (isotope) renal scans: MAG3 and DMSA

P1 · P2 **merged from 2**

Asked as: P1 LAQ "Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications" · P2 SAQ "Renal isotope scans and DMSA renal scan"

"Uro" is read as urology. "MAG 3" is taken to mean the Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.

Principles

- Nuclear medicine shows **function, not structure**. A radiotracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image. Planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)**, 3D, with CT (about 10 mm).
- **PET (positron emission tomography):** the positron annihilates with an electron, giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise and allows true quantitation (SUV, standardised uptake value); sharper (about 5 mm).

	Planar	SPECT	PET
Image	2D	3D, with CT (computed tomography)	3D, with CT or MRI (magnetic resonance imaging)
Resolution	About 20 mm	About 10 mm	About 5 mm
Quantitation	Semiquantitative	Semiquantitative	SUV
Main use	Renal function	Problem-solving	Cancer

- **^{99m}Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **^{18}F:** about 110 minutes, lowest positron energy, so the sharpest PET image. ^{68}Ga comes from a generator; ^{11}C (20 minutes) needs an on-site cyclotron.

Applications in urology (overview)

Area	Tracer	Use
Kidney	^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3, -DMSA (dimercaptosuccinic acid)	GFR (glomerular filtration rate), split function, obstruction, scars, transplant
Bone	^{99m}Tc-MDP (methylene diphosphonate)	High-risk prostate cancer staging; low specificity
Prostate	**PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine	PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90%
Therapy	^{177}Lu-PSMA-617; radium-223	mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021)

Area	Tracer	Use
FDG (fluorodeoxyglucose) PET	Glucose analogue trapped in cells	**Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma)
Renal mass	Sestamibi SPECT; girentuximab PET	Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab
Adrenal	^{123}I-MIBG (metaiodobenzylguanidine), ^{68}Ga-DOTATATE, FDG	Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2)
Sentinel node	^{99m}Tc-nanocolloid plus blue dye	Penile cancer, impalpable nodes
Scrotum	^{99m}Tc-pertechnetate	Torsion cold, epididymitis hot
Reflux	Direct radionuclide cystography	VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram)
Infection	^{111}In-labelled leucocytes; ^{67}Ga	Occult abscess; not in transplant

Renal tracers

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

Feature	^{99m}Tc-DTPA	^{99m}Tc-MAG3	^{99m}Tc-DMSA
Handling	Glomerular filtration only	Protein-bound; tubular secretion, high extraction	Filtered, then retained in proximal tubules (megalin, cubilin); tubular cortical binding
Study	Dynamic	Dynamic	**Static cortical**, at 2 hours (P2: 2–4 hours)
Main use	**GFR**; captopril study; renovascular	**Split function, obstruction (first choice)**, drainage; transplant	**Pyelonephritis and scars**
Renal failure	Poor images in renal failure	Good images in renal failure	Delay imaging
Dose	5–10 mCi	5–10 mCi	5 mCi adult; 50 µCi/kg child

MAG3: principle

- **^{99m}Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA.
- So it gives **good images even with poor GFR**, lower background and lower cost. **Agent of choice for split function and obstruction.**

Dynamic and diuretic renography (MAG3): method and interpretation

- **Preparation:** good hydration; bladder catheter for outlet obstruction or neurogenic bladder, and in children; cap a nephrostomy. Dose 185–370 MBq (5–10 mCi) intravenous bolus in adults.

- **Perfusion phase** (first minute): frames every 1–2 seconds; the kidney appears within seconds of the aorta (renal curve follows aortic upstroke).
- **Function (uptake) phase:** regions of interest with background subtraction. **Split function (DRF, differential renal function) = area under each curve at 1–3 minutes**; **normal 45–55%** (45:55 to 55:45). Normal MAG3 peak at 3–5 minutes; bladder by 10–15 minutes.
- **Drainage phase:** a rising curve needs furosemide. Upright or post-void images test gravity drainage.
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.
- **Diuretic renography ("well-tempered" diuretic renogram):** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full; separates a patulous system from obstruction. Timing varies (F+20 classic, F+15, F0, F−15), so the report must always state when it was given.

T½ (half-clearance time) after furosemide	Meaning
<10 minutes	Not obstructed
10–20 minutes	Indeterminate
>20 minutes	Obstructed

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, not obstructed, washes out (falls) with diuretic; IIIb equivocal.

MAG3 uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR; split function before nephrectomy.

MAG3 pitfalls

- Poor function, dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation (capacious pelvis) and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

DMSA renal scan

Principle

- **^{99m}Tc-DMSA** is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.
- Images at **2–4 hours** (P1: 2 hours; later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT**, which is slightly more sensitive.
- Radiation about 1 mSv.

What it measures

- **DRF:** from geometric-mean counts; normal 45–55%. Better than MAG3 for ectopic, horseshoe or duplex kidneys and very poor function.

- **Cortical defects:** photopenic areas.

Indications

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR and recurrent UTI; top-down approach (DMSA first, VCUG only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Deciding whether to reconstruct or remove a poorly functioning kidney.

Acute pyelonephritis versus scar

	Acute pyelonephritis	Scar
Timing	Within 1 week of fever (sensitivity falls by 1 month)	**≥6 months** after infection
Defect	Ill-defined, may be multiple	Sharp, with cortical thinning
Kidney size	Normal or swollen	Small or contracted (often small)
Repeat scan (at about 6 months)	Resolves (about 85%)	Persists

Goldraich scar grading

Type	Finding
1	No more than 2 scarred areas
2	More than 2 scars, normal tissue between
3	Generalised damage, small kidney
4	End-stage, shrunken kidney, little or no uptake

Limitations

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

Other renal applications (special studies)

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this support, so uptake falls. Repeat off ACE (angiotensin-converting enzyme) inhibitors: improvement **≥10%** off ACE inhibitors means renovascular hypertension is likely. Largely replaced by CT angiography.
- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Shows urine leak without iodinated contrast.
- **Radionuclide (nuclear) cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower radiation dose than VCUG, but little anatomy. So VCUG first; nuclear cystography for follow-up of reflux.

Infection imaging

- **^{111}In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. ^{99m}Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** ^{67}Ga-citrate (non-specific, slow).

Oncology

- **FDG (^{18}F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
- **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
- Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
- Poor in prostate cancer and clear cell RCC.
- **Prostate: PSMA PET** is preferred (NCCN) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan (^{99m}Tc-MDP) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** ^{123}I-MIBG and ^{68}Ga-DOTATATE for phaeochromocytoma; ^{99m}Tc-nanocolloid sentinel node biopsy in penile cancer; ^{99m}Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

Therapy (theranostics)

- **^{177}Lu-PSMA-617:** beta emitter for mCRPC after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy).
- Poor anatomy; PET is costly; results depend on technique.

Diagrams to draw

- MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window; renogram curve types I (normal), II (obstructed, rising), IIIa (dilated non-obstructed, falls after furosemide) and IIIb.
- Gamma camera chain: patient, collimator, crystal, photomultiplier, image.
- Posterior DMSA image: normal kidney and a scarred kidney with polar defects.

Key points to remember:

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 minutes obstructed, <10 minutes not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h; gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

LAQ 6 Energy sources in urology and laparoscopic surgery, including bipolar energy

P2 · P3 · P4 **merged from 3**

Asked as: P3 LAQ "Therapeutic energy sources in urology (including laparoscopic energy devices)" · P4 LAQ "Basic energy modalities in urological surgery, including bipolar energy" · P2 SAQ "Energy sources in laparoscopic urology"

Introduction and principles

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- All haemostatic devices achieve haemostasis (seal) by **denaturing protein**, by heat (current, light) or mechanical vibration.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

Classification

Group	Modalities
Electrical	Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation
Acoustic	Ultrasonic dissector, SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy)
Direct contact	Cryoablation, water vapour (Rezum), waterjet (Aquablation)

1. Electrosurgery: physics

- Alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat, too fast to stimulate nerve or muscle.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats (burns); a large pad disperses it.
- **Monopolar:** generator, active electrode, patient, dispersive pad, back to generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two electrodes (jaws) on the same instrument; less power, less spread, no dispersive pad (the patient is not in the circuit).
- **Electrocautery** (heated wire) differs: no current enters the patient.

Waveform	Character	Effect
Cut (50–80 W)	Continuous, low voltage	Cell water flash-boils; rapid boiling, clean incision, little char
Coagulation (30–50 W)	Interrupted (about 6% duty cycle), high voltage	Desiccation, deeper spread (up to 3 mm)

Waveform	Character	Effect
Fulguration (spray)	Higher voltage, arcs (sparks) across air	Non-contact surface haemostasis; surface charring of oozing beds
Blend	Cut with interruptions	Cutting plus haemostasis

Bipolar energy in detail

- Radiofrequency current (about 500 kHz). Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

	Monopolar	Bipolar
Circuit	Loop to patient to pad	Active and return poles on the instrument
Irrigant	Glycine 1.5%, sorbitol–mannitol	Normal saline
TUR (transurethral resection) syndrome	Yes	Theoretically eliminated
Capacitive coupling, pad burns	Yes	No

- **Endoscopic uses:**
- **Bipolar TURP (transurethral resection of prostate):**
  - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
  - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
  - *True bipolar:* dual loops.
  - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
- **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
- **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
- **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.
- **Laparoscopic and open uses:** bipolar forceps for pinpoint haemostasis; vessel sealers; prostatic pedicles, cystectomy.
- **Bipolar safety:** safer near pacemakers; touching tines complete the circuit without tissue. Obturator reflex reduced, not abolished.

Electrosurgery in endoscopic urology: present status

- **TURP:** still the reference BPH (benign prostatic hyperplasia) operation. Monopolar TURP needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** in saline matches monopolar efficacy with better safety and avoids TUR syndrome.
- **TURBT:** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.

- **Obturator reflex:** lateral-wall TURBT stimulates the nerve, causing adduction and perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

Laparoscopic energy devices

Device	Principle	Lateral spread	Notes
Monopolar (hook, scissors)	Tip to patient to return pad	**5–7 mm**	Versatile; stray-current risk
Bipolar	Current only between jaws	1–6 mm	No pad; no capacitive coupling
Vessel sealer (LigaSure, EnSeal, Caiman)	Bipolar current + jaw compression fuses collagen and elastin; impedance feedback stops output; built-in blade	1–3 mm	Seals vessels **≤7 mm** in one application only
Ultrasonic (Harmonic)	Blade vibrating at **25–55 kHz** (P4: about 55.5 kHz); friction and cavitation denature protein; no current through the patient	**≈1 mm**	Safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel
Argon beam	Monopolar current through argon, non-contact	Superficial	Liver, spleen, raw surfaces; **gas embolism**

- Vessel sealers are routine in laparoscopic and robotic surgery.
- **Also know:** lasers (CO_2, KTP [potassium titanyl phosphate], holmium) now rarely used laparoscopically; laparoscopic RFA (target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses; robotic HotShears, bipolar Maryland, Vessel Sealer.

Electrosurgical injury and safety

Mechanism	Prevention
Direct (inadvertent) activation out of view	Activate only with tip in view
Direct coupling: active electrode touches another metal instrument	Whole metal tip visible
Capacitive coupling: current induced through intact insulation (metal cannula in plastic retainer)	Never mix metal and plastic ports (avoid hybrid trocars)
Insulation failure	Inspect insulation; **AEM (active electrode monitoring)**
Thermal spread to ureter, nerves or bowel	Bipolar near vital structures
Return pad burns; surgical fire (with oxygen)	Pad on muscle, away from implants; **REM (return electrode monitoring)** detects poor pad contact

- **AEM** detects insulation failure and capacitive coupling.
- Unseen bowel injuries present late (days) as perforation or peritonitis.
- Use lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view; bipolar near vital structures and in pacemaker patients.

2. Ultrasonic energy

- **Ultrasonic dissector:** see the device table; little lateral spread, but the blade stays hot.
- **Intracorporeal ultrasonic lithotripter:** vibrating probe with suction of fragments.

3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity gives stimulated emission and a **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.
- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.

Laser	nm	Penetration	Use and present status
KTP/LBO (potassium-titanyl-phosphate/lithium triborate; GreenLight)	532	0.8 mm	Prostate photovaporisation (haemoglobin); can continue anticoagulation
Nd:YAG (neodymium:yttrium-aluminium-garnet)	1064	10 mm	Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH
TFL (thulium fibre laser)	1940	0.08 mm	Stones: beats low-power holmium, equals high-power; less retropulsion; enucleation (ThuLEP, thulium laser enucleation of prostate)
Tm:YAG (thulium:YAG)	2000	0.2 mm	Vaporesection, enucleation
Ho:YAG (holmium:YAG)	2120	0.31 mm	Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation

- Safety: eye protection for all; irrigant heating; fibre damage.

4. Acoustic energy and lithotripsy

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Shock waves break brittle stones by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones.
- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser; pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam (1500–10,000 W/cm^2) is absorbed and heats a rice-grain focus to necrosis; transrectal prostate. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy** (cavitation liquefaction), **burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

5. Ablative energies (thermal ablation of tumours)

Modality	Mechanism	Key points and present status
Cryoablation	Argon expansion (Joule–Thomson); **below −40 °C, two freeze–thaw cycles**; visible ice ball	Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate
RFA	375–900 kHz current; ionic agitation; target 60 °C; char at 150 °C	Small renal mass; oncological results similar to cryo; heat sink near vessels
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Renal mass; hotter, larger zones, less heat sink
HIFU	Focused ultrasound absorbed, causing necrosis	See acoustic energy
IRE	High-voltage microsecond DC (direct current) pulses open permanent membrane pores; non-thermal	Spares vessels and ducts; focal prostate; good continence and potency; trials ongoing

- **Cryo mechanism:** extracellular ice, then cell dehydration, then intracellular ice, then membrane rupture on thawing; ischaemia and apoptosis follow.
- **IRE mechanism:** electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.
- **Guidelines:** AUA 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

6. Light and chemical energy

- **PDT:** photosensitiser plus light gives singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard. EMDA for mitomycin (see above).

7. Prostate (BPH): newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal saline waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

8. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours).
- **Radioligand therapy** (^{177}Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

Diagrams to draw

- Monopolar vs bipolar circuit (monopolar loop-to-pad beside bipolar loop with local return), with cut and coagulation waveforms.
- Laser penetration depths compared, with cryoprobe ice ball.

Key points to remember:

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome; bipolar TURP equals monopolar efficacy with fewer complications.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink; thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

LAQ 7 LASERs in urology: holmium, thulium, KTP and newer laser technologies

P4 **merged from 4**

Asked as: P4 LAQ "LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment)" · P4 SAQ "Holmium laser" · P4 SAQ "Thalium laser in urology" · P4 SAQ "KTP*** laser in urology"

*"Thalium" is taken to mean thulium (spelling error). "KTP***" is taken to mean the KTP (potassium titanyl phosphate) "GreenLight" laser; the asterisks are ignored.*

Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

Lasers used

Laser	Wavelength (nm)	Chromophore	Penetration	Main uses
Ho:YAG	2120 (Ch 19 quotes 2100; often quoted as 2100 nm)	Water	0.3–0.4 mm (other sources: 0.4 mm; 0.31 mm)	Stones, HoLEP, incisions, tumours
TFL	1940	Water (peak)	<0.1 mm (other sources: about 0.2 mm; 0.08 mm; chapter values 0.08–0.2 mm)	Dusting, enucleation
Tm:YAG	~2010 (other sources: 2013)	Water	0.2 mm	Vaporesection, enucleation
KTP/LBO	532	Haemoglobin	0.8 mm	PVP
Diode	940–1470	Water + haemoglobin	0.5–5 mm	Prostate vaporisation
Nd:YAG	1064	Neither strongly	Up to 10 mm	Coagulation of tumours
CO_2	10,600	Water	~0.1 mm	Skin and penile lesions

Holmium:YAG laser

Physics

- Solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).

- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

Mechanism on stones

- Long pulses (>250 µs) act **photothermally**, not by shock waves: long pulses heat the stone until it melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate. It is the gold standard for stones and enucleation.

Settings

Mode	Settings	Effect
Fragmenting	0.6–1.0 J × 6–10 Hz	Pieces for basketing; more retropulsion
Dusting	0.2–0.3 J × 40–80 Hz	Fine dust; "popcorning" in a calyx
HoLEP enucleation	2.0 J × 40–50 Hz	Cutting along the capsule

- **Retropulsion** rises with pulse energy (not frequency) and falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

Uses

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate).
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele, calyceal diverticulum neck.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.
- Drawback: steep HoLEP learning curve.

Thulium lasers

Types and physics

	Tm:YAG (thulium:yttrium-aluminium-garnet)	TFL (thulium fibre laser)
Wavelength	2013 nm (P4 LAQ: ~2010 nm)	**1940 nm** (water absorption peak)
Source	Diode-pumped solid crystal	Diode-pumped thulium-doped silica fibre
Output	Continuous wave (now also pulsed)	Pulsed or continuous
Water absorption	High	About 4× holmium

- Optical penetration is the shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG.
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

Thulium:YAG (2013 nm)

- Originally continuous wave (0.2 mm penetration, carbonisation a drawback): ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (laser enucleation, 2010; mostly blunt dissection; size-independent like HoLEP).
- **Pulsed Tm:YAG:** newer pulsed solid-state version of the continuous-wave thulium laser for photothermal lithotripsy and enucleation.

TFL (1940 nm) for stones

- **Diode-pumped silica fibre** laser near the **water absorption peak** (about **4× holmium absorption**).
- **Low peak power, long pulses, very high frequency** (up to about 2000 Hz [P4 LAQ: above 2000 Hz]; holmium about 120 Hz); pulse energy as low as 0.025 J. This gives photothermal fine "dusting" and **minimal retropulsion**.
- **Thin fibres** (50–150 µm; P4 LAQ: down to about 150 µm) give better irrigation and scope deflection. Small, quiet, compact unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone); settings ≤40 Hz are often enough. "Paint, do not drill."
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate, with less retropulsion.
- Concerns: efficacy on all stone types; thermal injury at high power.

Thulium for prostate

- Techniques: ThuVARP, ThuVAP, ThuVEP, ThuLEP, ThuFLEP (thulium fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP.
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

Other thulium uses: en bloc bladder tumour resection; UTUC ablation; incision of strictures.

KTP/LBO (GreenLight) laser

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

Platforms

Generation	Power
KTP	80 W
LBO HPS (High Performance System)	120 W
LBO XPS	**180 W**; 750 µm MoXy fibre (metal tip, fluid cooling)

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

PVP (photoselective vaporisation of prostate) technique

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

Indications

- AUA: prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

GOLIATH trial

- RCT, 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

Complications

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

Other KTP uses: laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

Applications by area

1. Stones

- **Ho:YAG** is the gold standard and breaks all stone types (see above). **TFL:** higher water absorption and low peak power give finer dust and less retropulsion.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

2. BPH

- **HoLEP:** endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate. The only endoscopic procedure superior to TURP. Less bleeding, shorter catheter time; safe on anticoagulants.
- **PVP (KTP/LBO 120–180 W):** haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue. **GOLIATH:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA: HoLEP, ThuLEP or PVP for **high bleeding risk**.

3. Incisions

- Laser urethrotomy has no advantage over cold knife.

- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

4. Urothelial tumours

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC:** biopsy, then ablate by ureteroscopy:
- Ho:YAG: <0.5 mm, haemostatic.
- Nd:YAG: 5–6 mm coagulation, so stricture risk.
- Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

5. Penile and genital lesions

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- CO_2 is too superficial for T1 (recurrence up to 50%). CO_2 or Nd:YAG for genital warts.

Newer generation laser technologies (new lasers for stone treatment)

- **Aims:** less retropulsion, finer dust, faster work.
- **1. High-power holmium (60–120 W; other sources: 100–120 W machines) with pulse modulation:** high frequency allows **dusting** (0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.
- **2. Pulse modulation (Moses technology and similar):** the first part of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone. Moses technology splits each pulse in two: the first opens the channel, the second strikes the stone. Less retropulsion; may shorten HoLEP.
- **3. TFL (1940 nm):** see above; ThuFLEP for prostate.
- **4. Pulsed Tm:YAG (2013 nm):** see above; it also does lithotripsy.
- **Others:** thin fibres; suction sheaths; AI (artificial intelligence) settings; 180 W GreenLight KTP/LBO (532 nm) vaporisation; diode lasers; burst wave lithotripsy (investigational).

Holmium versus TFL

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Water absorption	High	About 4× higher
Penetration	0.31 mm	0.08 mm
Peak power	High	Low
Retropulsion	More	Minimal
Fragments	Small	Smaller
Fibre	≥200 µm	≥150 µm

Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.
- TFL settings ≤40 Hz often enough.

Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.
- Check fibres; never fire inside the scope. Guard against fire and thermal (urothelial) injury.
- **Heat follows total power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; use low power, pauses, high flow, cool irrigation and an **access sheath**; urothelial damage begins above about 43 °C.

Diagrams to draw

- Water and haemoglobin absorption curves marking 532, 1064, 1940, 2013 and 2120 nm.
- Vapour bubble (Moses channel) between fibre tip and stone.
- Side-firing fibre at 70° vaporising lateral lobe.

Key points to remember:

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium: 2120 nm; water chromophore; pulsed; photothermal; Moses effect: fibre close and perpendicular.
- Holmium settings: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
- HoLEP size-independent, superior to TURP.
- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- ThuLEP is size-independent and equal to HoLEP.
- KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
- Nd:YAG for PeIN; eyewear for everyone.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

LAQ 8 Describe IVU & its applications in Urology.

P1

Type: LAQ (25 marks)
Source: 2021-Summer Paper 1, Q2

Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage must be seen.

Contrast media

Type	Example	Feature
High-osmolar ionic	Diatrizoate	More reactions; obsolete for IV use
Low-osmolar non-ionic (standard)	Iohexol, iopamidol	Fewer reactions
Iso-osmolar non-ionic	Iodixanol	High-risk renal patients

Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

Technique: film sequence

Film	Timing	Shows
Scout (control) KUB (kidney–ureter–bladder)	Before contrast	Calcification, bones, gas, technique
Contrast	Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV	—
Nephrogram / nephrotomogram	About 1 min (60–90 s)	Renal size, outline, parenchyma
Pyelogram	5 min	Calyces; then apply **ureteric compression**
Compression film	10 min	Well-distended calyces and pelvis
Release (full-length) film	About 15 min	Whole ureters

Film	Timing	Shows
Bladder and **post-void** films	20–30 min	Bladder outline, residue, lower ureters
Delayed films	Up to 24 h	Level of obstruction; **no excretion at 24 h = non-functioning kidney**
- **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis.		
- Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma.		

Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

Nephrogram patterns

Pattern	Causes
Absent	Renal artery occlusion, agenesis, avulsion
Delayed, increasingly dense	**Acute obstruction**, renal artery stenosis
Striated	Acute pyelonephritis, obstruction
Rim (cortical rim) sign	Renal infarction (capsular collaterals)
Bilateral persistent dense	Hypotension, ATN (acute tubular necrosis)

Applications in urology

Condition	IVU features
Calculi	Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy)
Acute obstruction	Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture
PUJ (pelviureteric junction) obstruction	Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation
Duplex system	Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**"
Ureterocele	"**Cobra head**" (spring onion) at the bladder
Horseshoe kidney	Low, medially rotated; lower calyces point medially
Retrocaval ureter	"Reverse J" or **fish-hook** at L3
Medullary sponge kidney	"**Paintbrush**" papillary striations
Papillary necrosis	**Ring sign**, "lobster claw" calyces

Condition	IVU features
Genitourinary TB (tuberculosis)	**Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy
Urothelial tumour	Irregular filling defect, non-visualised calyx
Retroperitoneal fibrosis	Medial deviation and tapering of ureters with hydronephrosis
Pelvic lipomatosis	Pear-shaped bladder
Schistosomiasis	Ureteric striations, calcified bladder
- **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient.	
- **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula.	

Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min (maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

Diagram to draw: normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

LAQ 9 Robotics in urology (including robotic vs standard laparoscopy)

P4

Asked as:

- Role of robotics in urology — *Summer 2015 Paper IV Q1 (LAQ)*
- Discuss Robot and its role in urology. — *Winter 2017 Paper IV Q1 (LAQ)*
- Robots in urology — *Practice Set 4 Paper IV Q2 (LAQ)*
- Robotic vs std lap surgeries — *Practice Set 3 Paper IV Q3b (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)); "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; first paediatric robotic procedure **2002**; robotic RPLND (retroperitoneal lymph node dissection) 2006.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
Standard (2000)	Three arms
S (2006) / Si (2009)	High definition, fourth arm / dual console for training
Xi (2014)	8-mm camera in any port, finer arms, side docking (docks from either side), no stirrups
SP (2018)	All instruments through one 2.5-cm cannula
da Vinci 5 (2024)	Force feedback

- Patents expired in 2019 (after patent expiry), so other platforms appeared (e.g. **Hugo, Versius** and the low-cost Indian **SSI Mantra**).
- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

Robotic vs standard laparoscopy: comparison

Feature	Standard laparoscopy	Robotic
Vision	Mostly 2D; assistant holds camera	**3D**, 10× magnified, surgeon controls camera
Instruments	Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement	**EndoWrist, 7 DOF**, 90° articulation; intuitive movement
Precision	Tremor transmitted	**Tremor filtration, motion scaling**
Ergonomics	Standing, awkward	Seated, less fatigue
Suturing	Hard; steep learning curve	Easy; shorter learning curve
Haptic feedback	Present (limited)	Absent (force feedback only in newest system)
Port placement	Flexible	Ports 8–10 cm apart; docking time
Bedside	Surgeon at patient	Surgeon remote; emergency undocking delays
Cost	Low, reusable	High capital; instruments limited to about 10 uses
Failure	—	Robot failure about 0.4%

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost; cost-effective only at high volume
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- **Retzius-sparing** and **"hood"** techniques preserve continence structures and give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.

- Versus laparoscopy: laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy (robotic vs laparoscopic gives): **fewer conversions**, **shorter warm ischaemia**, less (smaller) eGFR (estimated glomerular filtration rate) loss/fall and shorter stay; complications and margins similar. So more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.
- ICG (indocyanine green) guides selective clamping.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions (less transfusion).
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications. Continent diversion fell (11% in iROC).

4. Other procedures

Procedure	Role of the robot / why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children and infants (small working space: about 1 L vs 6 L in adults); similar success to laparoscopy; easier suturing
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy; adrenalectomy suits the retroperitoneal approach
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) / fistula repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

- **Radical nephrectomy, adrenalectomy:** no outcome advantage over laparoscopy; robot only adds cost.
- Outcomes depend more on the **surgeon** than on the platform.

Choosing robot vs laparoscopy

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children; another source: work at 10–12 mm Hg) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery increases venous return and cardiac output, and raises intracranial and intraocular pressure.
- The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

Complications specific to robotics

Problem	Details
Steep Trendelenburg (~22–25°)	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury (up to ~6%)	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position; fasciotomy if >30 mmHg with symptoms
Thermal bowel injury / bowel burns	Insulation failure, capacitive coupling; presents late
Robot malfunction	About 0.4% (another source: <1%); half found before the patient enters; rarely forces conversion
Port-site hernia	Close larger port sites; transverse incisions cut it from 5.4% to 0.4% after RARP

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulator (simulators), animal and cadaver labs, then proctored cases (proctoring). **FRS (Fundamentals of Robotic Surgery)** curriculum improved skills in an RCT. Five-day mini-residencies also help. Dual console and simulators for training. Proficiency comes after about 40 cases, but outcomes keep improving (outcomes improve) with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; da Vinci SP (single port); force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency; latency limits it).
- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs.
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement; EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

Key points to remember:

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

LAQ 10 Flexible endourological instruments, digital and current ureteroscopes

P4

Asked as:

- Briefly outline flexible endourological instrument and its role in management of urological disease — *Practice Set 1 Paper IV Q1 (LAQ)*
- Digital flexible ureteroscopes — *Summer 2019 Paper IV Q3c (SAQ)*
- Current ureter scopes. — *Summer 2022 Paper IV Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

Summer 2022 Q3d taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.

Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.
- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

Classification of current ureteroscopes

Type	Key features
Rigid rod-lens	Larger calibre; largely historical
Semirigid	**Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece
Flexible fibreoptic	Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots)
Flexible digital	"Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head
Single-use flexible	Digital; LithoVue (2016) first; now several makers
Ureteral resectoscope	11.5 Fr; largely replaced by laser

Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.

- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

Components

Component	Key features
Optics	**Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable
Deflection	Lever → control wires through movable metal rings → wire fixed at the tip (deflection mechanism: lever pulls wires). "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection
Working channel	Single channel shared by irrigation and instruments, so instruments reduce flow and deflection
Shaft	Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access

- Flexible ureteroscope parts: optics, deflection mechanism (lever pulls wires), working channel.

Sensor technology (digital)

	CCD (charge-coupled device)	CMOS (complementary metal-oxide semiconductor)
Heat and power	More	Less (runs cooler)
Cost, speed	Higher, slower	Cheaper, faster
Status	Phased out	Current standard

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

Types

Instrument	Specifications	Main use
Flexible cystoscope	14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120°	Office diagnosis and minor therapy
Flexible ureteroscope	Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists	Kidney and upper ureter
Flexible nephroscope (or flexible cystoscope used through the tract)	Larger calibre; too wide for the UPJ (ureteropelvic junction)	Clearing calyces during PCNL (percutaneous nephrolithotomy)

Deflection and channel (flexible ureteroscope)

- Active deflection **≥270° both directions** (another source: **>270° both ways** in one plane); enhanced primary deflection eases lower-pole access / reaches the lower pole (ureteroinfundibular angle up to about 175°).
- Working channel **3.6 Fr**; dual-channel scopes exist.

Fibreoptic vs digital (digital versus fibreoptic)

Feature	Fibreoptic	Digital
Image	Honeycomb pattern; black dots as fibres break; degrades	Sharper, larger (magnified), stable image; no focusing or white balance
Size / Tip	Smallest tips, tapered	Larger (about 8.5 Fr), flatter; more access-sheath use
Outcome / Operative time	Reference; longer	Shorter operating time, equal (similar) stone-free rate (SFR)
Durability	Fragile; over-rotation damages fibres; repair about every 15 cases	**Not better** in later comparisons; not clearly better in trials
Processor	Platform-agnostic	Proprietary

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.
- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.
- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

Single-use scopes (single-use vs reusable)

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale (reason): reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Best value in low-volume centres and for **large lower-pole stones**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Cost advantage depends on case volume; environmental impact is similar to reusables. Most centres use a **hybrid fleet**.

Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.
- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

Role in urological disease

Area	Role
Lower tract	Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract
Stones	RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe
UTUC (upper tract urothelial carcinoma)	Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding)
Obstruction	Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm
Diagnosis	Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval
PCNL	Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery)

Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; **fire laser only when the fibre tip is seen touching the stone** (fire only with the fibre tip seen on the stone); leave the bladder empty to avoid buckling.
- **Sterilisation:** sterilise with ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS (ureteroscopy):** ergonomics, less radiation to surgeon.
- **Also know:** suction (access) sheaths, further miniaturisation.

Diagram to draw: cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires (scope tip cross-section showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection).

> **Key points to remember:**
>
> - Working channel plus active deflection made flexible scopes therapeutic.
> - fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
> - Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
> - Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
> - Semirigid: two channels; below the iliac vessels.
> - Pressure sensors and AI are the newest developments.
> - Golden rule: never fire unless the fibre tip is seen on the stone.

LAQ 11 Natural orifice endoscopic surgery.

P4

Type: LAQ (25 marks)
Source: 2022-Summer Paper 4, Q1

Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).

Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

History (milestones)

Year	Event
2002	Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu)
2004	Transgastric peritoneoscopy in pigs (Kalloo)
2006	NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers
2008–2010	Hybrid, then pure, transvaginal nephrectomy in humans

Routes of access

Route	Features	Urological use
Transvaginal (posterior colpotomy)	Commonest; easy closure under vision; large specimens pass; women only	Nephrectomy, donor nephrectomy, specimen extraction
Transgastric	Upper abdomen reachable; gastrotomy closure and leak are the main risk	Experimental (peritoneoscopy, renal surgery in animals)
Transvesical	Sterile urine; bladder dome port; small calibre; closure difficult	Peritoneoscopy (animal, Lima), transvesical access to pelvis
Transurethral	Urologist's familiar route	Experimental transurethral prostatectomy in cadavers
Transcolonic/transanal	Faecal contamination risk	Rarely urological

Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with

real clinical urological use.

- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.
- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.
- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.

- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

Also know: LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

Diagram to draw: sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

> **Key points to remember:**
> - NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
> - Transvaginal extraction removes the most painful wound.
> - LESS: one incision, lost triangulation, more hernias.
> - Barriers: closure, infection, orientation, platform; no proven benefit yet.

LAQ 12 Laparoendoscopic single-site (single-port) surgery: LESS

P4

Asked as:

- Laproendoscopic single site surgery: LESS — *Winter 2023 Paper IV Q1 (LAQ)*
- Single port lap. Surgeries in urology — *Practice Set 2 Paper IV Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered as a long answer in 2023-Winter Paper 4, Q1)

Read as laparoendoscopic single-site surgery (LESS), i.e. single-port (single-site) laparoscopic and robotic surgery in urology. NOTES (natural orifice transluminal endoscopic surgery) is covered in 2022-Summer Paper 4, Q1.

Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter (pass) through **one skin incision**, usually hidden in the umbilicus (umbilical) or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Sub-types (variants): transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS); purpose-built **SP (single-port) robot**.

Access: ports and devices

Option	Description
Several low-profile trocars	Placed close together through one fascial incision (the future extraction incision)
Multichannel ports	TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it)
da Vinci SP entry guide	One 25–27-mm cannula carrying three articulating (wristed) 6-mm instruments + one flexible articulating camera

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera (instruments lie parallel to the camera).
- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
- **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).
- **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
- A 2-mm needlescopic instrument or extra 5-mm port (an extra needlescopic port) without hesitation.
- **Robotic platforms** restore wristed movement.

Robotic single-site surgery

Platform	Features
da Vinci Single-Site (Si/Xi)	Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control
da Vinci SP (FDA [Food and Drug Administration] approval 2018)	Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions

- SP needs **10–25 cm between (from) cannula and (to) target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children (paediatric).
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient (patients), small or benign lesion (lesions), experienced surgeon.
- **Relative contraindications (avoid in):** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial (RCT, randomised controlled trial):** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion (otherwise equal); **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.

- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined (limited) to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

Complications

- All laparoscopic complications (usual laparoscopic injuries: access injury, vascular and bowel injury, CO_2 effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.
- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems / instruments (restore triangulation).

Diagram to draw: umbilical multichannel port with articulating (crossed) instruments crossing at the fascia (chiral) versus a triangulated multiport diamond (layout).

> **Key points to remember:**
> - LESS = one incision; LESSCAR (2008) standardised the term.
> - Core problem: lost triangulation; solved by articulating or curved instruments and robots.
> - Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
> - da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Short answer questions (SAQ)

SAQ 1 Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa

P1 **merged from 3**

Asked as: P1 SAQ "Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa" · P1 SAQ "Andrew Novick." · P1 SAQ "Christian Chaussey"

Names read as Prof. Harold Hopkins ("Herald"), Dr Charles Huggins, Prof. Sir Roy Calne ("Caine"), Dr D. D. Gaur ("Gaue"), Hari Asopa, Andrew C. Novick and Christian Chaussy ("Chaussey"). The question is taken to ask for each person's contributions to urology. Novick is answered mainly through nephron-sparing surgery, and Chaussy through the history and principles of SWL.

Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**.

Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

Dr Charles B. Huggins (1901–1997)

- Canadian-born American urologist, University of Chicago.
- **1941, with Clarence Hodges:** castration (or oestrogen) relieved far-advanced disease and caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens made it worse. This proved that **prostate cancer is hormone (androgen) dependent**.
- Used serum **acid phosphatase** to monitor response: the first tumour marker.
- **1945:** bilateral adrenalectomy for castration-resistant disease (relapse after castration, suspecting adrenal androgens); modest, brief responses; morbidity high.
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer: the basis of hormonal cancer therapy.
- **Legacy:** basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.

Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

Hari S. Asopa (Indian urologist)

1. Asopa dorsal inlay urethroplasty (2001)

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.
- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

2. Asopa hypospadias repair (1970s)

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

Andrew C. Novick

Who he was

- A **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

Indications for PN

Category	Situations
Imperative	Bilateral RCC (renal cell carcinoma); solitary functioning kidney
Relative	Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC
Elective	cT1a (≤4 cm) with a normal other kidney; now larger selected tumours

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak. - Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

Christian Chaussy (Munich): extracorporeal shock wave lithotripsy

Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the**

Dornier HM1. They reported it in the *Lancet* in 1980.

- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

Principles

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

Energy source	Feature
Electrohydraulic (HM1, HM3)	Underwater spark, ellipsoid reflector; most effective; electrode wear
Piezoelectric	Ceramic elements; long life, little pain; small focal zone, low energy
Electromagnetic (modern)	Coil and membrane with lens; consistent; dominant today

Mechanisms of fragmentation

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

Practical points

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

Diagrams to draw

- Conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps).
- Dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).
- PN steps: clamp, ice slush, excision, closure, renorrhaphy.
- Ellipsoid reflector with spark at F1 and stone at F2.

Key points to remember:

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966; 1945 adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery; PN indications imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer; negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1; HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

SAQ 2 Bacteriuria in pregnancy and antibiotic policy

P1 · P3 **merged from 2**

Asked as: P1 SAQ "Bacteriuria in pregnancy." · P3 SAQ "Baceriuria and use of antibiotic policy in pregnancy"

Read as: bacteriuria in pregnancy and the antibiotic policy for it.

Definition

- **ASB (asymptomatic bacteriuria):** ≥10^5 CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms (≥10^2 CFU/mL on a catheter sample).

Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%; P3: about 2–7%), but ASB **persists and ascends**. Higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

Why pregnancy matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB leads to pyelonephritis in 15–45%** (historical), mostly in the third trimester (60–75% of cases). Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces pyelonephritis, preterm delivery and low birth weight.

Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.

- Treat all ASB found in pregnancy.

Antibiotic policy

Safe	Restricted or avoided
β-lactams (penicillins such as amoxicillin; cephalosporins such as cephalexin): throughout	**Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis
Fosfomycin: lower tract (cystitis) only	**TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st trimester only if no alternative.
- **ASB and cystitis:** culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (ampicillin plus gentamicin, ceftriaxone or aztreonam) until afebrile, then oral (cephalexin) to complete ≥14 days (P3: 10–14 days).
- No improvement by **72 h: renal ultrasound** (stone, obstruction, abscess).

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015: close surveillance equally effective).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy and consider prophylaxis.

Diagram to draw

- Right hydroureter compressed at the pelvic brim.

> **Key points to remember:**
> - Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
> - Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
> - β-lactams safe throughout; no fluoroquinolones or tetracyclines.
> - Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

SAQ 3 Evidence-based medicine and critical appraisal

P1 · P4 **merged from 2**

Asked as: P1 SAQ "Evidence-based medicine and critical appraisal of research papers" · P4 SAQ "Describe evidence based medicine."

"quaLliy" is taken to mean quality, i.e. critical appraisal.

Definition

- **EBM (evidence-based medicine):** "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Critical appraisal: three core questions

1. Is it valid? (internal validity)

- **PICO question**; suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

Bias	Meaning
Selection	Groups differ at entry
Confounding	Third factor (comorbidity, frailty) linked to exposure and outcome; needs risk adjustment
Performance, detection, attrition	Unequal care, outcome assessment or dropouts
Lead-time, length-time; stage migration (Will Rogers)	Screening or better tests make survival look longer
Publication	Negative studies unpublished

- **Pitfalls in appraisal:** selection, confounding, lead-time, length-time, stage migration (Will Rogers) and attribution bias.

2. What are the results?

- Effect: RR (relative risk), **ARR, NNT = 1/ARR**.
- Precision: **95% CI**; p value; **sample size and power** (usually 80%; α 0.05).
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does); validated outcome tools (validated PROs [patient-reported outcomes]); Clavien–Dindo grading.

3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted by EAU [European Association of Urology])

Level	Type of evidence / design
1a	Systematic review or meta-analysis of RCTs
1b	At least one RCT

Level	Type of evidence / design
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports, case series
4	Expert committee reports or opinion

- **Grades:** A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU).
- AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Reporting checklists and quality tools

Design	Checklist
RCT	**CONSORT** (Consolidated Standards of Reporting Trials)
Systematic review / meta-analysis	**PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses)
Observational	**STROBE** (Strengthening the Reporting of Observational studies in Epidemiology)
Diagnostic accuracy	**STARD** (Standards for Reporting Diagnostic accuracy studies)
Surgical innovation	**IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study)

- **Also know:** Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest; Clavien–Dindo; validated PROs; Donabedian model.

Limitations of EBM

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw

- Evidence pyramid and the three overlapping circles of EBM.

> **Key points to remember:**
> - Sackett 1996: best evidence + clinical expertise + patient values.
> - 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
> - Level 1a = systematic review of RCTs; GRADE: strong or weak.
> - NNT = 1/ARR; beware lead-time, length-time and confounding bias.
> - Three questions: valid, results, applicable.
> - Randomisation, concealment, blinding, ITT, follow-up >80%.
> - CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

SAQ 4 Microscopic / asymptomatic haematuria and urinary biomarkers

P2 · P3 **merged from 2**

Asked as: P3 SAQ "Asymptomatic haematuria and urinary biomarkers" (Asymptomatic Hematuria, Summer 2016 Paper III Q3d; Urinary biomarkers in diagnostic evaluation of hematuria, Summer 2021 Paper III Q3f) · P2 SAQ "Microscopic Hematuria" (Practice Set 1 Paper 2 Q3e; also 2016 Summer Paper 3 Q3d)

Definitions and epidemiology

- **MH (microscopic haematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **A dipstick alone is not enough**: myoglobin, semen and bacterial peroxidases give false positives.
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy in about 3% of MH (over 85% of these are bladder cancer) and in 14–26% of GH (DETECT I, IDENTIFY).
- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.

Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma) and RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell.
- Anticoagulants do not explain haematuria; evaluate as usual.

Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**, urine culture.
- Suspected UTI: culture, treat, then **repeat urinalysis**; persistent MH means full evaluation.
- Glomerular or renal picture (dysmorphic RBCs, casts or proteinuria suggest a glomerular source): nephrology **and** urological evaluation.

AUA/SUFU 2020 risk stratification

	Low (all)	Intermediate (any)	High (any)
Age	Women under 50, men under 40	Women 50–59, men 40–59	60 or over
Smoking (pack-years)	Never or under 10	10–30	Over 30
RBCs/HPF	3–10	11–25	Over 25

	Low (all)	Intermediate (any)	High (any)
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT (computed tomography) urography**

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move a patient from low to intermediate.
- CT contraindicated: MR (magnetic resonance) urography, or retrograde pyelograms.
- Persistent MH after a negative ultrasound: CT or retrograde imaging.
- Hereditary RCC risk (VHL (von Hippel–Lindau)): upper tract imaging in any stratum.
- White-light cystoscopy. Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (graded by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology with normal cystoscopy: random and prostatic urethral biopsies, upper tract washings.

Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

Marker	Principle	Sens / spec	Note
NMP22 (nuclear matrix protein 22)	Nuclear protein; lab or point-of-care	0.69/0.77	False positives: inflammation, stones
BTA (bladder tumour antigen) stat / TRAK	Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay)	0.64/0.77	**False positive with blood** (factor H in serum)
UroVysion FISH (fluorescence in situ hybridisation)	Gain of chromosomes 3, 7, 17; loss of 9p21	0.63/0.87	After atypical cytology
ImmunoCyt	Antibodies to CEA (carcinoembryonic antigen) + two mucins	0.78/0.78	Highest sensitivity; operator dependent
Cxbladder	RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs	about 0.82/0.85	Needs validation

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive and less specific than cytology; they miss small low-grade tumours.

Role of markers in haematuria (guidelines)

- **AUA/SUFU 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

Follow-up

- Negative evaluation: consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up (haematuria pathway), with markers shown as adjuncts.

> **Key points to remember:**
>
> - MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
> - Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.
> - Cytology: specific for high grade/CIS, insensitive for low grade.
> - BTA is falsely positive with blood.
> - No markers in initial microhaematuria work-up; none replaces cystoscopy.

SAQ 5 Hemostasis in laparoscopic surgery

P3 · P4 **merged from 2**

Asked as: P3 SAQ "Hemostasis in laparoscopic surgery." · P4 SAQ "Hemostasis in lap surgery"

Principles

- Devices seal by **denaturing protein**. Plan from imaging.
- Keep a **haemorrhage tray** ready: laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent.

Effect of pneumoperitoneum

- CO_2 at 12–15 mmHg **tamponades venous bleeding**, hiding it.
- Raise the pressure briefly for venous bleeding: 25 mmHg (P4: 20–25 mmHg).
- Before exit, **lower the pressure and inspect** the operative site and each port to unmask hidden bleeders and port-site bleeding: about 5 mmHg (P4: 5–10 mmHg).

Energy devices

Device	Lateral thermal spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure, direct coupling
Bipolar	1–6 mm	Current only between the jaws

Device	Lateral thermal spread	Note
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current; blade >200 °C, cools slowly
Argon beam	Superficial	Raw surfaces; gas embolism risk

Clips

- **Titanium clips:** no diathermy near them (they slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhages, the manufacturer **contraindicated Hem-o-lok on the living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

Staplers

- Endo GIA through a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

Sutures and haemostatic agents

- Intracorporeal or barbed sutures.
- Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing bleeding

- **Venous injury** (algorithm): 1. Raise pneumoperitoneum pressure (25 mmHg; P4: 20–25 mmHg). 2. Suction to identify the bleeding point. 3. Pledget tamponade. 4. Sealant. 5. Extra 5-mm port for suturing or Satinsky. 6. Hand-assist. 7. Convert.
- Keep the patient well filled (gas embolism).
- **Arterial:** minor bleeding, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

> **Key points to remember:**
> - Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
> - LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
> - Hem-o-lok contraindicated on the donor renal artery.
> - Never staple across clips; grey or white loads for vessels.

SAQ 6 Newer generation CT scan

P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3a

Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

Evolution

Generation	Feature
Helical (~1989)	Continuous table movement, volumetric data
Multidetector (1998 → 320-row)	Sub-millimetre slices, 3D reconstruction
Dual-source/dual-energy (2005)	Two energies: material analysis
Photon-counting (US approval 2021)	Energy-sorts each photon: less noise, dose and metal artefact

Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

Low-dose CT

Protocol	Dose
Conventional abdomen/pelvis	10–20 mSv
Low-dose	<3–4 mSv
Ultra-low-dose	1–2 mSv
- Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first.	

Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

Contrast phases and protocols

Phase	Timing	Use
Arterial	20–30 s	Renal arteries, donor anatomy, active bleeding
Corticomedullary	30–70 s	Hypervascular (clear cell) tumours, renal vein
Nephrographic	90–120 s	Most sensitive for renal masses
Excretory	7–10 min	Collecting system, ureter, urine leak

Phase	Timing	Use
- **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions.		
- **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts.		
- **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma).		
- **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold.		

Diagram to draw: Timeline: axial → helical → multidetector → dual-source → photon-counting.

> **Key points to remember:**
> - Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
> - DECT: uric acid stones, virtual unenhanced, iodine maps.
> - Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

SAQ 7 History of Laparoscopy

P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3b

Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

Optics and access

Year	Pioneer	Milestone
1806	**Bozzini**	**Lichtleiter**, candle-lit light conductor
1877	Nitze	Distal electric light and lenses
1901	**Kelling**	Insufflated abdomen of a **dog**
1910	**Jacobaeus**	First human series; "laparoscopy"
1938	**Veress**	Spring-loaded needle
1966	**Hopkins**	**Rod-lens system**
1970s–80s	**Semm**	Automatic insufflator; operative laparoscopy
1970s	Hasson	Open access
1987	Mouret	Video cholecystectomy

Year	Pioneer	Milestone
- **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; CO_2 became the standard insufflant.		

Urological milestones

Year	Pioneer	Milestone
1976	**Cortesi**	Impalpable testis, first urological use
1990–91	**Clayman**	**First laparoscopic nephrectomy**; nephroureterectomy 1991
1992	**Gaur (India)**	**Balloon retroperitoneal access**
1995	**Ratner**	Live-donor nephrectomy
1997	Schuessler	Laparoscopic radical prostatectomy (slow, not adopted)
2000	Guillonneau & Vallancien	Standardised prostatectomy
2000	**da Vinci**; **Binder**	Robot cleared in US; first robotic prostatectomy
2000s	**Menon**	Vattikuti Institute Prostatectomy
2007	Raman; Rane	Single-port nephrectomy

Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

Diagram to draw: Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

> **Key points to remember:**
> - Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
> - Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
> - da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

SAQ 8 Foley's (urethral) catheters, guidewires and DJ stents with their complications

P1

Asked as:

- Foley's catheter. — *Summer 2016 Paper I Q3e (SAQ)*
- Urethral catheters, guidewires and DJ stents — *Practice Set 18 SAQ Q3f (SAQ)* (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)

- Complications if DJS — *Summer 2021 Paper I Q3c (SAQ) — Taken as complications of a DJ (double-J) ureteric stent.*

Type: SAQ (10 marks)

Foley's / urethral catheters

History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

Design

- **Size: Fr (French) = 3 × diameter in mm.** Adults (adult) usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: (male) meatus 24 Fr (narrowest point), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use (inflate balloon with) **sterile water**, only after urine flows at the hub; volume is printed on the port.
- **Lumens:** one (intermittent, specimen); two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria (stiff wall resists collapse). Each lumen reduces (shrinks) drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation; better urethral tolerance); PVC (polyvinyl chloride); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.

Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first (**suspected urethral injury → RUG first**).
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon (puncture with a wire).

CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

Guidewires

Feature	Details
Uses	Access, safety wire, straightening ureter, guiding dilation and stents
Construction	Stainless-steel core with outer wrap; **nitinol** core is kink-resistant
Coating	PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft
Size	0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm
Tip and stiffness	Straight, angled, J; standard to super-stiff (straightens tortuous ureter)

DJ (double-J) stents: introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder; adult usually 6 Fr, length by height. Consent should cover its symptoms and complications.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

Classification of DJ stent complications

Timing	Complications
At insertion	**Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed)
Early	**Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding)
Late	**Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula**

Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

Encrustation and forgotten stent

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on removal**, renal loss.
- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

Stent failure

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

Prevention

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

Diagram to draw: cross-section of a three-way Foley; DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

> **Key points to remember:**
>
> - Foley balloon catheter: designed for post-prostatectomy haemostasis.
> - Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
> - Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
> - CAUTI 5–10% per day; remove early; change the catheter when treating.
> - Hydrophilic wire to pass, PTFE wire to stay (safety).
> - DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
> - Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
> - Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

SAQ 9 Proteinuria

P1

Asked as:

- Proteinuria — *Summer 2017 Paper I Q3f (SAQ)*
- Proteinuria — *Summer 2021 Paper I Q3d (SAQ)*

Type: SAQ (10 marks)
(also answered in 2017-Summer Paper 1, Q3f)

Definition

- Normal protein excretion is **<150 mg/day** (another source: **80–150 mg/day**) (albumin <30 mg/day). The filter (glomerulus) blocks (holds back) albumin and the proximal tubule reabsorbs filtered small proteins.

- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

Detection and quantification (KDIGO, Kidney Disease: Improving Global Outcomes, categories by ACR)

Category	ACR (albumin-to-creatinine ratio)	PCR (protein-to-creatinine ratio)	24-h protein
A1 normal–mild	<30 mg/g (<3 mg/mmol)	<150 mg/g (<15 mg/mmol)	<150 mg
A2 moderate	30–300 mg/g	150–500 mg/g	150–500 mg
A3 severe	>300 mg/g (>30 mg/mmol)	>500 mg/g	>500 mg
- **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check. Alternatives: PCR, 24-h urine.			
- **Dipstick** (tetrabromophenol blue) detects albumin only (mainly albumin), from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** detects (precipitates) all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma.			

Types

Type	Mechanism	Amount	Example
Glomerular (commonest)	Leaky filter	Often >2 g; albumin ≈70% (about 70%)	Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis), amyloid
Tubular	Failed reabsorption of small proteins	<2 g (seldom >2–3 g); mainly globulins	Fanconi syndrome, heavy metals, drugs
Overflow	Excess plasma protein	Variable (0.3–2 g)	**Myeloma** (Bence Jones), myoglobin, haemoglobin

By timing	Features
Transient	Fever, exercise, heart failure; commonest in children
Orthostatic	Young men, upright only, <1 g; benign (normal first-morning sample). Test with a split (supine overnight vs upright day) collection
Persistent	Mostly glomerular; needs full work-up

Evaluation

`Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise → persistent → quantify (ACR, PCR or 24-h protein) → creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells / RBCs, casts), ultrasound → glomerular pattern (>2 g, albumin) → nephrology ± renal`

biopsy; C3, C4, ANA if systemic → globulins → electrophoresis, immunofixation (myeloma)

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest (suggests) medical renal disease; urological bleeding rarely raises protein much (adds little protein).
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer (favour) nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

Diagram to draw: glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

> **Key points to remember:**
> - Normal <150 mg/day; nephrotic >3.5 g/day.
> - ACR A2 30–300, A3 >300 mg/g.
> - Glomerular, tubular, overflow; orthostatic proteinuria is benign.
> - Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

SAQ 10 Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion

P1

Asked as:

- Health related quality of life instruments (HRQOL) in urology. — *Summer 2018 Paper I Q3c (SAQ)*
- Assessment of quality of life after urinary diversion — *Practice Set 3 Paper I Q3c (SAQ)*

Type: SAQ (10 marks)

Definition

- **HRQOL (HRQoL, health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother; the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It differs from a **symptom score**, which only counts symptoms.
- Must be **patient-reported**; doctors underestimate dysfunction (urinary, sexual and bowel dysfunction).

Types of instruments

Type	Measures	Examples	Limitation
Generic	Domains common to all diseases; allow comparison across diseases	**SF-36** (Short Form-36), SF-12, **EQ-5D** (EuroQol five-dimension; 5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile	Insensitive to diversion-specific issues
Cancer-specific	Cancer effects	**FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer core questionnaire)	General cancer domains
Disease-specific	Domains of one disease; more responsive	See below	
Bladder-cancer-specific		**FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index)	Weight symptoms of one diversion, so biased in cross-diversion comparisons
Neobladder-specific		IONB-PRO (23 items)	Not usable across diversions

Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-BI, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.
- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

Psychometric properties (choosing a tool)

- A good tool is **reliable** (Cronbach α >0.7), **valid** in that population and **responsive** to change.

Property	Meaning	Threshold
Internal consistency	Items agree	Cronbach α >0.7 (>0.9 = redundant)

Property	Meaning	Threshold
Test–retest reliability	Stable over time	r >0.7
Content (face) validity	Right items present	Expert judgement
Criterion validity	Agrees with gold standard	r ≥0.7
Construct validity	Convergent and divergent	Correlations
Responsiveness	Detects real change	**MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

Quality of life after urinary diversion

Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

Diagram to draw: Pyramid: symptoms → function → bother → overall HRQOL. Bar sketch: body image (neobladder better) versus urinary function (conduit better).

Key points to remember:

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

SAQ 11 PET scans in urology.

P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3e

Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: ^{18}F 110 min, ^{68}Ga 68 min, ^{11}C 20 min (on-site cyclotron).

Tracers

Tracer	Mechanism	Main use
^{18}F-FDG (fluorodeoxyglucose)	Glucose analogue trapped after phosphorylation (Warburg effect)	Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma
PSMA (prostate-specific membrane antigen) ligands (^{68}Ga-PSMA-11, ^{18}F-DCFPyL, ^{18}F-PSMA-1007, rhPSMA-7.3)	Bind cell-surface PSMA	Prostate staging and recurrence
^{11}C/^{18}F-choline	Cell membrane synthesis	Prostate recurrence (older)
^{18}F-fluciclovine	Amino acid transport	Prostate recurrence
^{18}F-NaF (sodium fluoride)	Bone turnover	Bone metastases

Uses by organ

Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.

- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **^{177}Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

Testis

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

Bladder and UTUC

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

Kidney

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **^{89}Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

Penis

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

Diagram to draw: Positron annihilation with two 511-keV photons hitting a detector ring.

> **Key points to remember:**
> - proPSMA: PSMA PET 92% vs 65% accuracy.
> - Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
> - FDG poor in prostate and clear cell RCC.
> - VISION: ^{177}Lu-PSMA-617 in mCRPC.

SAQ 12 Recist Criteria.

P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3a

Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

Lesions at baseline (RECIST 1.1)

Category	Definition
Measurable lesion	Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers

Category	Definition
Measurable node	**≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal
Non-measurable	Small lesions, blastic bone metastases, effusions, ascites
Target lesions	Up to **5 (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded
Non-target	All other lesions, recorded as present or absent

Response categories

Response	Target lesions
CR (complete response)	All disappear; nodes <10 mm
PR (partial response)	**≥30% fall** in sum versus **baseline**
PD (progressive disease)	**≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion**
SD (stable disease)	Neither PR nor PD

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors)); bone poorly assessed; observer variability.

Modifications

Criteria	Use
iRECIST (2017)	Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression**
Choi	Density fall counts as response
PCWG3 (Prostate Cancer Working Group 3)	Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST
PERCIST	PET (positron emission tomography) metabolic response

Diagram to draw: number line of change in sum: −30% (PR), +20% (PD).

> **Key points to remember:**
> - RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
> - PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
> - iRECIST for immunotherapy; PCWG3 for prostate.

SAQ 13 Outline algorithm for difficult catheter placement.

P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3b

Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

Algorithm (Campbell Fig. 15.6)

First attempt fails ├─ Prostatic obstruction suspected │ → 18-Fr coudé, tip (marker) pointing up │ → fails → flexible cystoscopy └─ Not suspected (stricture likely) → smaller catheter → fails → hydrophilic glidewire → fails → flexible cystoscopy At cystoscopy / with wire in bladder: ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm │ → super-stiff wire → serial dilatation (1 Fr above catheter size) │ → Councill (open-tip) catheter over wire └─ False passage → pass under vision (scope anterior) → wire → Councill over wire Wire cannot be passed / dense stricture / urethral disruption → suprapubic catheter (ultrasound-guided percutaneous, or open)

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

Special situations

Situation	Approach
AUS	Deactivate; smallest catheter 12–14 Fr; remove early
Obese or female	Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide
Buried penis	Infrapubic pressure; flexible cystoscopy
Neobladder, continent pouch	Never force; scope + wire; suprapubic only with imaging guidance
PUV (posterior urethral valves) neonate	6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy

Also know: filiforms, balloon dilators; complications: false passage, haematuria, stricture.

Diagram to draw: coudé catheter following the bulbar-prostatic curve with tip up.

Key points to remember:

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

SAQ 14 Indications for outpatient cystourethroscopy.

P1

Type: SAQ (10 marks)
Source: 2024-Winter Paper 1, Q3e

Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

Diagnostic indications

Indication	Comment
Haematuria	All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk
Bladder cancer surveillance	NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma)
Recurrent UTI	Look for stone, diverticulum, stricture, fistula, foreign body
LUTS (lower urinary tract symptoms)	Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture)
Pain syndromes	IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow
Women	Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery
Others	Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body

Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

Diagram to draw: haematuria risk-group flowchart.

> **Key points to remember:**
> - Haematuria and NMIBC surveillance are the two commonest indications.
> - Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
> - Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

SAQ 15 MR urography

P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3a

Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

Techniques

	Static MRU	Excretory MRU
Sequence	Heavily T2-weighted; urine is bright	Gadolinium-enhanced T1
Contrast	None	GBCA (gadolinium-based contrast agent)
Depends on function	No	Yes; low-dose furosemide improves it
Best for	Dilated or obstructed system	Non-dilated system, excretion

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.
- Upper tract tumours and renal masses.

Advantages and limitations

Advantages	Limitations
No radiation; best anatomical detail	Cost, availability, long study (up to 1 hour)
Anatomy, function and drainage in one test	Sedation often needed under 6 years
As accurate as CTU (CT urography) for obstruction	Poor for stones; complex post-processing
Safe in iodinated contrast allergy	Pacemakers, metal implants, claustrophobia

Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

Diagram to draw: coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

> **Key points to remember:**
> - Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
> - Stones are signal voids; CTU preferred for urothelial cancer.
> - Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
> - NSF risk minimal with group II agents.

SAQ 16 Urine cytology

P1

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 1, Q3b

Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

Category	Criteria
Non-diagnostic	Inadequate
Negative for HGUC	N:C <0.5, or atypia explained by a benign cause
AUC (atypical urothelial cells)	N:C ≥0.5 and one feature
Suspicious for HGUC	N:C ≥0.7, at least two features, few cells (<5–10)
HGUC	N:C ≥0.7, at least two features, many cells
LGUN (low-grade urothelial neoplasm)	Fibrovascular cores; rarely diagnosable
Other malignancy	Squamous, adenocarcinoma, secondary

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).
- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

Diagram to draw: normal urothelial cell beside an HGUC cell with high N:C ratio.

> **Key points to remember:**
> - High specificity, low sensitivity; good for high grade and CIS.
> - Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
> - Positive cytology, normal cystoscopy: upper tracts, random biopsies.
> - Not for screening or initial microhaematuria.

SAQ 17 Patient positioning in lap urology

P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3a (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.

- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

Positions

Position	Used for	Set-up	Hazards
Steep Trendelenburg, low lithotomy or split legs	RARP (robot-assisted radical prostatectomy), cystectomy	Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30°	Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus
Modified flank (30–45°)	Transperitoneal nephrectomy, pyeloplasty	Back on gel rolls; upper arm across chest; wide tape; tilt to test	Pressure points, slippage
Full lateral (90°), table flexed	Retroperitoneoscopy, adrenalectomy	**Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows	**Rhabdomyolysis**; sciatic stretch
Prone	Posterior adrenalectomy	Pad torso, elbows, hips	Airway, face and eyes

Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

Diagram to draw: full flank position with axillary roll and table break.

> **Key points to remember:**
> - Nerve injury about 2.8%; brachial plexus is commonest at risk.
> - Never shoulder braces in steep Trendelenburg.
> - Axillary roll below, not in, the axilla.
> - Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

SAQ 18 Clinical trial methodology

P1

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 1, Q3a

Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

Phases

Phase	Purpose	Subjects
0	Microdose pharmacokinetics	Very few
I	Safety, dose, toxicity	20–80, often volunteers
II	Activity, efficacy signal	Patients, about 100–300
III	Comparative efficacy versus standard	Hundreds to thousands; basis of approval
IV	Post-marketing surveillance	General population

Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).

- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).
- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

Diagram to draw: CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

SAQ 19 Doppler in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3b

Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below 60°**.
- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

Modes

Mode	Shows	Use
Colour	Velocity and direction (red towards, blue away by convention)	Flow in kidney, testis, ureteric jets
Spectral (duplex)	Waveform at one point; velocities, RI	Penile and renal arteries
Power	Amplitude only; 3–5 times more sensitive, less angle-dependent	Low flow: torsion, prostate lesions

Applications

Organ	Findings
Scrotum	**Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis
Penis (after intracavernosal injection)	**PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment
Priapism	Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula

Organ	Findings
Kidney	Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy)
Transplant	Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI
Bladder	Ureteric jets (absent on the obstructed side)
Tumours	Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

Diagram to draw: spectral waveform marking PSV and EDV with the RI formula.

> **Key points to remember:**
> - RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
> - Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
> - Absent intratesticular flow = torsion, but explore on clinical grounds.

SAQ 20 Flow cytometry

P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3d

Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.
4. **Laser interrogation:** - **Forward scatter** = cell size. - **Side scatter** = internal complexity (granularity). - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations. - **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.

- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

Applications in urology

Area	Use
Bladder cancer	Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)**
Upper-tract urothelial cancer	Aneuploidy → poor long-term survival
Prostate cancer	Aneuploid tumours progress more after treatment than diploid
Neuroblastoma	**Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk
Urinalysis	Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria
Transplantation	**Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin
Male infertility	Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates
Lymphoma	Immunophenotyping of testicular or renal lymphoma

Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.
- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

Diagram to draw: DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

> **Key points to remember:**
> - Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
> - DNA index 1.0 diploid; aneuploidy = worse prognosis.
> - Flow crossmatch is more sensitive than CDC crossmatch.

SAQ 21 VCUG

P1

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 1, Q3c

Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm H_2O) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

Findings

- **VUR grade (International Reflux Study, 1981):**

Grade	Appearance
I	Ureter only, not dilated
II	Pelvis and calyces, no dilatation
III	Mild–moderate dilatation, minimal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.
- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

Diagram to draw: VUR grades I–V.

Key points to remember:

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

SAQ 22 Interventional radiology in urology and therapeutic angioembolisation

P1

Asked as:

- Therapeutic angioembolization. — *Practice Set 7 Paper I Q3e (SAQ)*
- Interventional urological treatment in urology — *Practice Set 8 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.

Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.
- **Requirements:** experienced team, resuscitation in the suite; unstable non-responders need surgery.

Non-vascular procedures

Procedure	Indications and points
PCN (percutaneous nephrostomy)	Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4%
Antegrade stenting	When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter
Antegrade pyelography, Whitaker test	Define strictures; pressure–flow test for obstruction
Percutaneous drainage	Renal or perinephric abscess >3 cm, urinoma, lymphocele
Image-guided biopsy	Renal mass biopsy

Vascular (endovascular) procedures: therapeutic angioembolisation

Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an abnormal vessel.
- **Superselective** embolisation of a segmental branch preserves most parenchyma.

Embolic agents

Type	Examples	Use
Temporary	Autologous clot; gelatin sponge (2–6 weeks)	Priapism, prostate, bladder bleeding
Permanent particles	PVA (polyvinyl alcohol), microspheres	Tumours, PAE (prostate artery embolisation)
Mechanical	Microcoils, vascular plugs	Pseudoaneurysm, AVF (arteriovenous fistula), varicocele
Liquid	NBCA (N-butyl cyanoacrylate) glue, absolute ethanol	Tumour ablation, recurrent priapism

Urological indications

Procedure	Indications
Renal angioembolisation	Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma)
Internal iliac (anterior division) embolisation	Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia)
PAE (prostate artery embolisation)	LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer
Varicocele embolisation	Coils or liquid sclerosants via femoral or jugular vein
High-flow priapism	Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40%
PTRA (percutaneous transluminal renal angioplasty)	Fibromuscular dysplasia; transplant renal artery stenosis

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.
- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

Complications

- Contrast nephropathy, access-site haematoma, coil migration.
- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; ED (erectile dysfunction) after priapism embolisation.

Diagram to draw: prone PCN through a posterior calyx along the Brödel line; renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

> **Key points to remember:**
> - PCN equals retrograde stent for decompressing an obstructed infected kidney.
> - Superselective coils: treatment of choice for pseudoaneurysm and AVF.
> - Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
> - Bladder bleeding: internal iliac anterior division; spare posterior division.
> - Post-embolisation syndrome is common and self-limiting.

SAQ 23 Tumour markers in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3d

Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

Testis (germ cell tumours)

Marker	Half-life	Source	Never raised in
AFP (α-fetoprotein)	5–7 days	Embryonal, yolk sac	Pure seminoma, choriocarcinoma
β-hCG (human chorionic gonadotrophin)	24–36 hours	Choriocarcinoma, embryonal; seminoma in ~15%	Yolk sac tumour
LDH (lactate dehydrogenase)	24 hours	Non-specific; reflects bulk	—

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

Diagram to draw: marker decline curve after orchidectomy using half-lives.

> **Key points to remember:**
> - PSA: organ-specific, not cancer-specific.
> - AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
> - S stage uses post-orchidectomy values.
> - Cytology: good for high grade only.

SAQ 24 Haemostatic agents in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3e

Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

Classification

Class	Examples	Mechanism / note
Passive (mechanical)	Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres	Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial)
Active	Topical thrombin	Converts the patient's fibrinogen to fibrin
Flowable	Gelatin matrix + thrombin (**Floseal**)	Fills irregular beds, swells
Fibrin sealants	Fibrinogen + thrombin (Tisseel, Evicel)	Final clotting step; works despite coagulopathy; also seals leaks
Synthetic sealants and glues	PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue)	Seal; adhesive
Systemic	Tranexamic acid, aminocaproic acid, desmopressin	Antifibrinolytic; desmopressin for uraemic platelets

Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

Intravesical agents (haemorrhagic cystitis)

Agent	Note
Alum 1% (50 g in 5 L)	First line; avoid in renal failure (aluminium toxicity)
Silver nitrate 0.5–1%	Chemical coagulation
Aminocaproic acid 200 mg/L irrigation	Thromboembolism and clot retention
Formalin 1–4%	Most effective; under anaesthesia; **cystogram first** to exclude reflux

Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.
- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

Diagram to draw: clotting cascade showing where passive agents, thrombin and fibrin sealant act.

Key points to remember:

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

SAQ 25 Sterilisation of urological instruments

P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3a

Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

Spaulding classification

Category	Contact	Requirement	Urological examples
Critical	Sterile tissue, vessels	Sterilisation	Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments
Semi-critical	Mucosa	HLD at least (sterilise if possible)	Flexible cystoscope, TRUS (transrectal ultrasound) probe
Non-critical	Intact skin	Low-level disinfection	Abdominal ultrasound probe

Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

Methods

Method	Conditions	Use and drawbacks
Steam autoclave	121 °C for 15 min (15 psi) or 134 °C for 3–4 min	Cheapest, most reliable; metal instruments, autoclavable optics
Dry heat	160 °C for 2 hours	Glass; rarely used
Ethylene oxide	Low temperature	Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration
Hydrogen peroxide gas plasma	Low temperature, short cycle	No toxic residue; not for cellulose; long narrow lumens limited
Peracetic acid (Steris)	Automated liquid, low temperature	Sporicidal; ureteroscopes

Method	Conditions	Use and drawbacks
Glutaraldehyde 2%	HLD in 20 min; sterilant after about 10 hours	Cheap; irritant, sensitiser; must rinse well
Gamma irradiation	Industrial	Single-use catheters, stents, sutures

Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

Diagram to draw: reprocessing cycle flowchart.

> **Key points to remember:**
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

SAQ 26 Ethics and informed consent in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3e

Four principles (Beauchamp and Childress, 1979)

Principle	Meaning
Autonomy	Patient's deliberated self-rule; basis of consent and confidentiality
Beneficence	Act for the patient's benefit, judged by the patient's values
Non-maleficence	Avoid harm (*primum non nocere*); requires training and honest risk data
Justice	Fair distribution of resources, benefits and risks

Principle	Meaning
- Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice).	
- **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law).	

Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented. - Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language. - Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.
- **Minors (<18 years):** parents consent; child gives assent.

Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.
- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

Diagram to draw: Jonsen four-box grid.

> **Key points to remember:**
> - Four principles: autonomy, beneficence, non-maleficence, justice.
> - Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
> - Commonest gap: not checking understanding.
> - Exceptions: emergency, incapacity (surrogate), minors.

SAQ 27 Operating room set-up

P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3f (see also 2021-Summer Paper 4, Q3a for operating room safety)

Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

Equipment layout by procedure

Setting	Arrangement
Endourology (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy))	Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready
Laparoscopic, upper abdomen	Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot
Laparoscopic, pelvic	30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite
Robotic	Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full CO_2 cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

Diagram to draw: theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

> **Key points to remember:**
> - Zones with one-way flow; positive pressure, about 20 air changes an hour.
> - Monitor directly ahead at eye level; open tray ready for laparoscopy.
> - Hypothermia triples wound infection (Kurz).
> - ALARA: time, distance, shielding.

SAQ 28 Amyloidosis in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3c

Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

Classification

Type	Protein	Cause
AL (amyloid light-chain; primary)	Immunoglobulin light chain	Plasma cell dyscrasia, myeloma
AA (amyloid A; secondary)	Serum amyloid A	Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis
ATTR (hereditary transthyretin)	Transthyretin	Autosomal dominant neuropathy
Dialysis-related	β2-microglobulin	Long-term dialysis
Localised	Usually AL	Deposits in one organ only

Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).

- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

Diagram to draw: Congo red stained fibrils with apple-green birefringence.

> **Key points to remember:**
> - Congo red, apple-green birefringence.
> - Localised bladder amyloid mimics cancer; biopsy then TUR.
> - Always exclude systemic AL disease.

SAQ 29 Collin's knife.

P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3b

*Taken to mean the **Collings knife**, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").*

Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

Uses (from the chapters)

Use	Technique point
TUIP (transurethral incision of prostate)	Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule
Bladder neck contracture (after TURP (transurethral resection of prostate) or simple prostatectomy)	Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence

Use	Technique point
Ureterocele incision (older children)	Low transverse full-thickness incision near the bladder floor to reduce reflux
PUV (posterior urethral valve) ablation	Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture)
Bladder diverticulum neck	Incise the narrow neck so the diverticulum drains
PCNL (percutaneous nephrolithotomy) tract	Incise scarred fascia when rigid dilators fail

Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

Diagram to draw: resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

> **Key points to remember:**
> - Knife-shaped resectoscope electrode for incision, not resection.
> - Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
> - Monopolar needs glycine or sorbitol; bipolar uses saline.

SAQ 30 APOGEE

P3

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 3, Q3c

Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.

Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

Diagram to draw: posterior mesh with two arms through the ischiorectal fossae.

> **Key points to remember:**
> - Apogee: transvaginal posterior/apical polypropylene mesh kit.
> - Exposure, dyspareunia and pain drove withdrawal.
> - FDA 2019 stopped transvaginal POP mesh sales.

SAQ 31 Urethral stents and metallic stents in urology

P4

Asked as:

- Urethral stents — *Summer 2015 Paper IV Q3d (SAQ)*
- Metallic stents in urology — *Practice Set 3 Paper IV Q3c (SAQ)*

- Urethral stents. — *Practice Set 4 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)

(urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

Definition and introduction

- A urethral stent is an endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.
- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

Classification (urethral stents)

Type	Examples	Features
Temporary	**Memokath** (nitinol), **Allium**	Not epithelialised (designed to avoid epithelial incorporation); removable; left 6–12 months
Permanent, epithelialising	**UroLume**	Urothelium grows over it; **withdrawn**
Biodegradable	Experimental	Dissolve

Classification of metallic stents (by site)

Site	Type	Example	Features
Ureter	Full-length coiled metal, double pigtail	**Resonance** (2006)	Tightly coiled alloy; changed every 12 months
Ureter	Segmental self-expanding mesh, bare or covered	Wallstent-type; covered nitinol (Allium)	Placed across malignant stricture; bare mesh allows ingrowth
Ureter	Thermo-expandable nitinol coil	Memokath 051	Removable
Prostate	Temporary thermo-expandable nitinol coil	**Memokath**	Not epithelialised; removable
Prostate, bulbar urethra	Covered self-expanding, removable	Allium	Left 6–12 months
Bulbar urethra	Permanent epithelialising mesh	**UroLume**	Withdrawn

Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

Urethral and prostatic stents

Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate) / unfit for surgery.
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: formerly (once) an alternative to sphincterotomy; now **abandoned** for complications.

Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed (with warm water) and softens when cooled (with cold water) for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

Results (outcomes)

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Memokath variable. Removable bulbar stents succeed only in small series. UroLume worked early (good early results, then ingrowth) but failed long term from **hyperplastic ingrowth** and restenosis, so withdrawn, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

Guidelines

- **AUA (American Urological Association)**: **no (urethral) stent if urethroplasty is considered** (planned); stents make reconstruction harder.

Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind** (temporary nitinol prostatic device): nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.
- **Also know:** biodegradable and drug-eluting stents (experimental).

Also know: a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

Diagram to draw: sagittal urethra with prostatic stent and bulbar stent distal to sphincter; prostatic Memokath ending above the external sphincter.

Key points to remember:

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

SAQ 32 Operating room safety : the role of surgeon.

P4

Type: SAQ (10 marks)
Source: 2021-Summer Paper 4, Q3a

Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

Phase	Key checks
Sign in (before induction)	Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL
Time out (before incision)	Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed
Sign out (before leaving theatre)	Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; FiO_2 (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.

8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

Diagram to draw: Swiss cheese model with aligned holes.

> **Key points to remember:**
> - WHO checklist: sign in, time out, sign out.
> - Universal Protocol: verify, mark, time-out.
> - Count discrepancy: ~100-fold risk of a retained item.
> - Fire triangle: surgeon owns ignition.

SAQ 33 Urological applications in nanotechnology.

P4

Type: SAQ (10 marks)
Source: 2022-Summer Paper 4, Q3a

Taken as "applications of nanotechnology in urology".

Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

Applications

Area	Application
Imaging and staging	**USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots
Sentinel node	Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84)
Diagnosis	Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels
Drug delivery	Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery
Focal ablation	**Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia
Irreversible electroporation	Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143)
Devices	Silver and other nanocoatings on catheters and stents against biofilm and encrustation

Area	Application
Tissue engineering	Electrospun nanofibre scaffolds that mimic extracellular matrix
Stone disease	"Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue

Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

Diagram to draw: nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

> **Key points to remember:**
> - 1–100 nm; enhanced permeability and retention allows tumour targeting.
> - USPIO MRI: benign nodes lose signal; metastatic nodes do not.
> - AuroLase gold nanoshells: photothermal focal therapy.

SAQ 34 Morcellation

P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3d

Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:** 1. Achieve good haemostasis first. 2. Pass a long nephroscope with a 5 mm channel. 3. Run two inflow lines fully open to keep the bladder distended. 4. Engage lobes with suction; cut only in mid-bladder, away from the wall. 5. Collect fragments in a trap for histology.
- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

	Intact extraction	Morcellation
Incision	Larger	Small, better cosmesis
Pain, time, stay	Similar	Similar
Pathology	Full staging	Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost
Seeding	—	Rare port-site recurrence

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

Diagram to draw: enucleated lobes in a distended bladder with morcellator suction.

> **Key points to remember:**
> - Morcellator made HoLEP practical; distend bladder, haemostasis first.
> - Morcellation loses stage information; histology still obtained.
> - FDA 2014: uncontained power morcellation spreads occult sarcoma.

SAQ 35 Stem cell therapy

P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3f (tissue engineering also in 2023-Winter Paper 1, Q1)

Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

Types

Type	Source	Comment
Embryonic	Blastocyst	Pluripotent; ethics, teratoma
Adult mesenchymal	Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord	Easy harvest; mainly paracrine
Muscle-derived (MDSC)	Skeletal muscle biopsy	Used for sphincter and detrusor
iPSCs (induced pluripotent stem cells)	Reprogrammed adult cells	Patient-specific; tumour risk

Type	Source	Comment
Urine-derived, spermatogonial	Urine; testis	Research

Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

Applications

- **SUI (stress urinary incontinence):**
- Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
- Aim: regenerate the striated sphincter, not just bulk it.
- Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
- After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.
- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.
- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

Diagram to draw: muscle biopsy → culture → periurethral injection into sphincter.

> **Key points to remember:**
> - Adult stem cells act mainly by paracrine signals.
> - SUI muscle-derived cells and ED stem cells are investigational.
> - Embryonic and iPSCs carry tumour risk.

SAQ 36 Advances in lap surgeries

P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3a (related: LESS in 2023-Winter Paper 4, Q1; energy sources

in 2015-Summer Paper 2, Q3d)

Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less CO_2 absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

2. Fewer and smaller incisions

Technique	Gain	Limitation
LESS (laparoendoscopic single-site surgery)	Cosmesis, less analgesia	No triangulation, clashing
NOTES (natural orifice transluminal endoscopic surgery)	No abdominal scar (hybrid transvaginal extraction)	No proven benefit
Mini-laparoscopy (2–3 mm instruments)	Less pain, cosmesis	Instrument flexibility
Hand-assisted	Tactile feedback, fast bleeding control	Larger wound

3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).
- **Also know:** augmented reality, artificial intelligence, telesurgery.

Diagram to draw: LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

SAQ 37 Artificial intelligence in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3d

Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

Development steps

`Data collection and labelling → training → internal validation → external validation → regulatory approval → use and monitoring`

Applications

Area	Examples
Imaging	CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv)
Pathology	Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021)
Endoscopy	Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature
Stones	Predict composition and stone-free outcome
Prognosis	Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy
Robotic surgery	Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy
Paediatrics	Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models
Others	Urodynamics interpretation; LLM chatbots for patient information

Advantages

- Fast big-data processing; consistency; decision support.

Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

Diagram to draw: neural network: input layer (image pixels), hidden layers, output (stone or no stone).

> **Key points to remember:**
> - Deep learning is part of ML, which is part of AI; CNNs for images.
> - CNNs: very high sensitivity for CT stones.
> - Paige Prostate: first FDA-authorised AI pathology tool (2021).
> - Limits: data, bias, black box, validation, liability.

SAQ 38 3D laparoscopy

P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3b

Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.
- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.

- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

Feature	2D laparoscopy	3D laparoscopy	Robot
Depth perception	No	Yes	Yes (magnified)
Wristed instruments	No	No	Yes
Cost	Low	Moderate	High

Diagram to draw: two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

> **Key points to remember:**
> - Two lenses, two images, polarised glasses: binocular disparity gives depth.
> - Main gain: faster, more accurate suturing; shorter learning curve.
> - Drawbacks: eye strain, headache, cost, glasses for all.
> - Robotic vision at low cost, but without wristed instruments.

SAQ 39 Telemedicine in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3c

Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).
- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

Types

Type	Method	Urological example
Synchronous	Real-time video or phone consultation	Stone, PSA (prostate-specific antigen) and post-operative follow-up
Asynchronous (store-and-forward)	Reports, images, bladder diaries sent for later review	Scan reports, uroflow traces
Remote monitoring	Apps and home devices	Bladder diary apps, home uroflowmetry

Type	Method	Urological example
Provider-to-provider	E-consults, telementoring, tele-tumour boards	Haematuria triage (Chapter 8)
Telesurgery	Remote robotic surgery	Experimental; limited by latency

Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

Diagram to draw: patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

> **Key points to remember:**
> - Telemedicine trades examination for access.
> - Best for follow-up, results and history-driven visits.
> - Limits: no examination, connectivity, elderly, medico-legal.
> - India: Telemedicine Practice Guidelines 2020; eSanjeevani.

SAQ 40 Simulators in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3d

Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

Types

Type	Examples	Strengths and limits
Bench / box trainer	Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney	Cheap, proven skill gains; limited realism
VR (virtual reality)	LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate)	Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel
Live animal	Pig	Real bleeding; ethics, cost
Cadaver	Fresh-frozen	Real human anatomy; scarce
Team (non-technical)	High-fidelity theatre crises	Communication and teamwork

Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.
- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

Diagram to draw: pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

Key points to remember:

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

SAQ 41 Virtual and augmented reality in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3e

Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

Workflow

`CT (computed tomography) / MRI (magnetic resonance imaging) -> segmentation of organ, tumour, vessels, collecting system -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR) -> registration to the live surgical view`

Applications

Use	Examples
Pre-operative planning	Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing
Intra-operative navigation	AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule
Education	VR anatomy, 360° operation videos; simulation, see Q3d
Patient counselling	Patients understand their anatomy and planned operation; better-informed consent
Paediatrics	VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38)

Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

Diagram to draw: CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

> **Key points to remember:**
> - VR immerses; AR overlays on reality.
> - Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
> - Biggest problem: registration on moving, deforming organs.
> - Evidence for outcomes still limited.

SAQ 42 3D printing in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3a

Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

Workflow

`Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces → STL (standard tessellation language) mesh → printer → post-processing`

- Takes hours to days; segmentation needs expertise.

Printing technologies

Method	Principle	Use
FDM (fused deposition modelling)	Melted plastic filament extruded	Cheap, rigid models
SLA (stereolithography)	Laser cures liquid resin	Fine detail, transparent models
Material jetting (PolyJet)	Jetted photopolymers, multi-colour	Tumour, vessels and calyces in different colours
SLS (selective laser sintering)	Laser fuses powder	Durable parts, devices
Bioprinting	Living cells in hydrogel "bio-ink"	Tissue engineering (see Practice Set 6 Paper 4, Q2)

Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.

2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

Diagram to draw: CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

Rapid Revision Sheet · Section 1

Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy (LAQ)

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main CA-AKI risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic.
- NAC is the antidote for paracetamol overdose; best within 8 h.
- Cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- CO_2 angiography: never above diaphragm.

Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision.
- Main indication for Hasson: previous surgery or adhesions.
- Hasson avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon in muscle causes hernia.
- Trocar = cannula + obturator.
- Bladeless trocars: fewer epigastric injuries and hernias.
- Balloon trocar seals and prevents retroperitoneal emphysema.
- CO_2 standard (soluble, non-flammable); helium in COPD; N_2O supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy (LAQ)

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.

- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

Tissue engineering in urology: principles, status and recent advances (LAQ)

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

Radionuclide (isotope) renal scans: MAG3 and DMSA (LAQ)

- ^{99m}Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- DMSA is the gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with ^{99m}Tc-nanocolloid.
- Sestamibi hot = oncocytoma; ^{177}Lu-PSMA-617 (VISION).

Energy sources in urology and laparoscopic surgery, including bipolar energy (LAQ)

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.

- Thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

LASERs in urology: holmium, thulium, KTP and newer laser technologies (LAQ)

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium: 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Holmium lithotripsy is photothermal; Moses effect: fibre close and perpendicular.
- Moses: two-part pulse through a vapour channel.
- Holmium: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: water peak, low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- ThuLEP is size-independent and equal to HoLEP.
- KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
- Nd:YAG for PeIN; eyewear for everyone.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

Describe IVU & its applications in Urology. (LAQ)

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

Robotics in urology (including robotic vs standard laparoscopy) (LAQ)

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

Flexible endourological instruments, digital and current ureteroscopes (LAQ)

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.

- Golden rule: never fire unless the fibre tip is seen on the stone.

Natural orifice endoscopic surgery. (LAQ)

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

Laparoendoscopic single-site (single-port) surgery: LESS (LAQ)

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa (SAQ)

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

Bacteriuria in pregnancy and antibiotic policy (SAQ)

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

Evidence-based medicine and critical appraisal (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Microscopic / asymptomatic haematuria and urinary biomarkers (SAQ)

- MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
- Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.
- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

Hemostasis in laparoscopic surgery (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Newer generation CT scan (SAQ)

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

History of Laparoscopy (SAQ)

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

Foley's (urethral) catheters, guidewires and DJ stents with their complications (SAQ)

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

Proteinuria (SAQ)

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion (SAQ)

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.

- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

PET scans in urology. (SAQ)

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: ^{177}Lu-PSMA-617 in mCRPC.

Recist Criteria. (SAQ)

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

Outline algorithm for difficult catheter placement. (SAQ)

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Indications for outpatient cystourethroscopy. (SAQ)

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

MR urography (SAQ)

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

Urine cytology (SAQ)

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Patient positioning in lap urology (SAQ)

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

Clinical trial methodology (SAQ)

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.

- ITT analysis; CONSORT flow diagram for reporting.

Doppler in urology (SAQ)

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

Flow cytometry (SAQ)

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

VCUG (SAQ)

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

Interventional radiology in urology and therapeutic angioembolisation (SAQ)

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

Tumour markers in urology (SAQ)

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

Haemostatic agents in urology (SAQ)

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

Sterilisation of urological instruments (SAQ)

- Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
- Autoclave 121 °C 15 min or 134 °C 3–4 min.
- Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
- Biological indicator: *G. stearothermophilus*.

Ethics and informed consent in urology (SAQ)

- Four principles: autonomy, beneficence, non-maleficence, justice.
- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

Operating room set-up (SAQ)

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Amyloidosis in urology (SAQ)

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Collin's knife. (SAQ)

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

APOGEE (SAQ)

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

Urethral stents and metallic stents in urology (SAQ)

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

Operating room safety : the role of surgeon. (SAQ)

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

Urological applications in nanotechnology. (SAQ)

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

Morcellation (SAQ)

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

Stem cell therapy (SAQ)

- Adult stem cells act mainly by paracrine signals.

- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

Advances in lap surgeries (SAQ)

- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

Artificial intelligence in urology (SAQ)

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

3D laparoscopy (SAQ)

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

Telemedicine in urology (SAQ)

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

Simulators in urology (SAQ)

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

Virtual and augmented reality in urology (SAQ)

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

3D printing in urology (SAQ)

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 2 · Oncology, DSD, Gender & Ageing

Long answer questions (LAQ)

LAQ 1 Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T)

P2 · P4 **merged from 2**

Asked as: P2 LAQ "Describe use and principles of immunology and Immunotherapy in Urological Oncology" · P4 LAQ "Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T."

Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette–Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).
- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG**, a tuberculosis vaccine, was among the first immune treatments for any cancer and is the original urological immunotherapy. It is non-specific; cancer vaccines aim for a **specific** anticancer response.

Principles of tumour immunology

Arm	Cells and role
Innate (rapid, pattern-based)	PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8))
Bridge	**Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node
Adaptive (specific, memory)	**CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells
Brake	**Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2). Without signal 2 the T cell becomes tolerant.
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.
- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

Immunoediting (three Es) and the rationale for immunotherapy in urology

- **Elimination** (small tumours destroyed), then **Equilibrium** (growth held), then **Escape** (clinical cancer).
- Escape is by loss of antigen or MHC, Treg expansion and checkpoint ligands. Vaccines try to restart the response.
- **Bladder:** inflammation-driven (schistosomiasis); CD8 infiltration is good; BCG success; high mutation load gives many neoantigens.
- **Kidney: CD8 paradox** (more CD8, worse outcome); rare spontaneous regression of metastases after nephrectomy; responses to IL-2 and interferon.
- **Prostate:** Treg-dominated and PD-L1 low, so checkpoint inhibitors fail; slow growth gives time to build a vaccine response; **organ-specific antigens** (PSA (prostate-specific antigen), PAP (prostatic acid phosphatase)).
- Vaccines have low toxicity and offer possible lasting memory.

BCG in NMIBC (non-muscle-invasive bladder cancer)

- Intravesical live attenuated mycobacteria.
- Mechanism (sequence): 1. Bacilli attach via **fibronectin**. 2. TLR2/4 activation → IL-8 and TNF-α. 3. Neutrophil wave → DC priming. 4. **Th1 CD4** response → memory.
- Th1 predicts success, Th2 failure. About half stay in remission for years.
- Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance; standard for high-risk NMIBC.

Cytokines and gene/viral therapy

- **IFN-α** and **high-dose IL-2**: durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak. Intravesical IFN-α: mixed results.
- **Nadofaragene firadenovec** (adenovirus carrying the IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) approved December 2022.
- **Cretostimogene**: oncolytic adenovirus with GM-CSF (granulocyte-macrophage colony-stimulating factor) gene.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

Cancer vaccines: mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, often GM-CSF, which activates DCs).
- Sequence: `Intradermal antigen + adjuvant → immature DC takes up antigen; adjuvant matures it → DC migrates to lymph node → peptide on MHC I/II = signal 1; B7 costimulation = signal 2 → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells → T cells traffic to tumour → apoptosis of antigen-bearing cells → memory T cells`
- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** the first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.
- **Resistance:** PD-L1 on tumour, Tregs, MDSCs, MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

Vaccine platforms

Platform	Example
Whole-cell (autologous or allogeneic)	Reniale (RCC (renal cell carcinoma)); GVAX (prostate)
Peptide/protein	IMA901
Dendritic-cell (ex vivo loaded)	Sipuleucel-T; AGS-003 (rocapuldencel-T)
Viral vector (poxvirus)	PROSTVAC-VF; TroVax (MVA-5T4)
Heat-shock protein–peptide	Vitespen (HSPPC-96)
DNA/mRNA, personalised neoantigen	Phase I–II trials

Vaccines by organ

Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, with low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. **No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (see note below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

Bladder cancer

- **BCG** (above); nadofaragene firadenovec, cretostimogene and N-803 (above).
- Peptide and neoantigen vaccines: experimental only.

Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern:** IMA901, AGS-003, PROSTVAC and GVAX all failed; **only sipuleucel-T succeeded in phase III.**

Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:** 1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells). 2. Cells incubated about 36 h with PA2024 at a central facility. 3. Infused intravenously about 3 days after collection. 4. **Three doses at 2-week intervals** (one-month course).
- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.

- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
- **OS 25.8 vs 21.7 months (HR (hazard ratio) 0.78)**, a gain of 4.1 months, despite placebo crossover.
- **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

Checkpoint inhibitors

Site	Use (trial)
BCG-unresponsive CIS	Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%)
Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC (upper tract urothelial carcinoma)	Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative
Metastatic UC (urothelial carcinoma)	Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab**
RCC	Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR 0.68)
Prostate	Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU (genitourinary) cancers; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies need hormone replacement.
- BCG toxicity: cystitis, BCG sepsis.

Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.
- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets; DNA/mRNA personalised neoantigen vaccines (phase I–II).

Diagrams to draw:

- T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.
- Sipuleucel-T flow: leukapheresis, PA2024 loading, reinfusion, T-cell priming.

> **Key points to remember:**
> - Three Es: elimination, equilibrium, escape; Th1 is antitumour.
> - BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
> - Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
> - IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
> - Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
> - CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

LAQ 2 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia

P3

Asked as:

- Management of DSD — *Summer 2019 Paper III Q3f (SAQ)*
- Evaluation and treatment of newborn with ambiguous genitalia — *Practice Set 4 Paper III Q2 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

DSD read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.

Introduction and definition

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital conditions with atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours (neonatal evaluation, first 24–48 h).

Classification (by karyotype)

Group	Main causes (examples)
46,XX DSD	**CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive, SRY+)
46,XY DSD	Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial; PAIS = partial androgen insensitivity); LH (luteinising hormone) receptor defect
Sex chromosome DSD	45,X/46,XY MGD (mixed gonadal dysgenesis); 45,X Turner; 47,XXY Klinefelter; chimerism

History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade (stage) I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven

otherwise (impalpable gonads suggest CAH). Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

Investigations (tests)

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (after 36 h; earlier values are falsely high); electrolytes and glucose daily. **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock); **salt-wasting appears around day 5–7** (usually after day 5–7), so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

Test	Meaning
AMH (anti-Müllerian hormone)	Present = functioning Sertoli cells
Testosterone, LH, FSH (follicle-stimulating hormone)	Testicular function (mini-puberty)
Androstenedione, DHEA-S (dehydroepiandrosterone sulfate)	Adrenal enzyme block
Testosterone:DHT (dihydrotestosterone) >20:1	5α-reductase deficiency
hCG (human chorionic gonadotropin) stimulation	No rise = no Leydig tissue or LH receptor defect

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH); then genitography, endoscopy, laparoscopy ± biopsy, genetic panels.
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels (genetic panels), microarray, exome. **No gonadectomy during diagnosis (while diagnosing).**

Diagnostic algorithm

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Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular; 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect) 46,XX + uterus + high 17-OHP -> CAH 46,XY, no uterus -> androgen synthesis/action defect 46,XY or 45,X/46,XY + uterus -> gonadal dysgenesis / MGD
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- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

Treatment

1. Emergency and medical management (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.
- **Puberty:** oestrogen or testosterone induction if gonads fail.

2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist; gender-neutral language. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

3. Sex of rearing (gender assignment)

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family wishes and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

4. Surgery: timing and consent

- Irreversible surgery is ideally deferred until the child can participate (consent); parents decide with the team after balanced counselling.
- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
- Clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often done in infancy (complication rates high).

5. Gonads (gonadal tumour risk)

Risk	Condition	Plan / management
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	Keep CAIS gonads until after puberty; ovotesticular: remove discordant tissue

Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

Diagram to draw: Prader stages I–V and the urogenital sinus with confluence level; karyotype-based DSD algorithm.

> **Key points to remember:**
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

LAQ 3 Discuss in brief, present concepts in various GU cancers

P4

Type: LAQ (25 marks)
Source: Practice Set 4 Paper 4, Q1

Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.

Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).
- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive** disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.
- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **^{177}Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.

- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

Diagram to draw: six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

> **Key points to remember:**
> - RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
> - UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
> - Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
> - Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

Short answer questions (SAQ)

SAQ 1 46,XX male

P1 · P2 **merged from 2**

Asked as: P1 SAQ "46-XX Male" · P2 SAQ "46 XX Male"

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition and classification

- **Testes and male genitalia in a person with a 46,XX karyotype.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006).
- First described by **de la Chapelle (1964)**. About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, together with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig cell testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	**Never (no AZF)**	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY** confirms; DSD gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).
- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support and bone density monitoring.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

> **Key points to remember:**
> - 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
> - Azoospermia always (AZF absent): no sperm retrieval.
> - Shorter than Klinefelter; FISH for SRY confirms.

SAQ 2 Female circumcision

P1 · P2 **merged from 2**

Asked as: P1 SAQ "Female circumcision" · P2 SAQ "Female Circumcision"

Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)"); the answer follows WHO (World Health Organization) definitions.

Definition and epidemiology

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- About **200 million** living women and girls are affected; about 3 million African girls a year. Mainly Africa, the Middle East and Asia (Indonesia, Malaysia); practised in some Indian communities ("khatna").
- Usually done in childhood, at 4–10 years, without anaesthesia, as a rite of passage; it predates Islam. Pharaonic type = infibulation.

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoral glans and/or prepuce
II (excision)	Removal of the clitoral glans and labia minora, with or without the labia majora
III (infibulation)	Narrowing of the vaginal opening (introitus) by cutting and joining the labia into a seal, with or without clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterisation

- The older Toubia scheme numbers differ: infibulation is Type III in WHO.

Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral or rectal injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, vaginal stone from urine stasis, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** retained menses, dysmenorrhoea, inclusion cysts, keloid, dyspareunia, apareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

Evaluation

- Sensitive, culturally respectful history; examination to classify the type; drawings or photographs for consent.
- Flow rate and residual urine; cystoscopy if fistula is suspected.

Management

- **Deinfibulation (type III):** anterior **midline division of the scar** under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn (subcuticular closure of each edge). High success.
- **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
- **Never reinfibulate** (illegal in the UK).
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited evidence and follow-up.

Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

Diagram to draw: vulva showing types I–III and the deinfibulation incision line.

Key points to remember:

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

SAQ 3 Familial / hereditary urological malignancies

P1 · P2 **merged from 2**

Asked as: P1 SAQ "Familial malignancies in urology" (Familial malignancies in Urology, Practice Set 2 Paper I Q3b; Familial malignancy in urology, Practice Set 8 Paper I Q3f) · P2 SAQ "Hereditary urological malignancies."

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts
HPRC	MET (7q31) oncogene	Multiple bilateral papillary (type 1, type 1-like)	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive**, often solitary papillary; collecting duct	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (folliculin, 17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3% (P2: 1–3%)	Angiofibromas, cortical tubers

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

Management of familial kidney tumours

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations.

- **Exception: HLRCC**: excise promptly, with wide margins.
- TSC AML: everolimus at 3–5 cm (P1; another source: if >3 cm); embolisation or excision (partial nephrectomy) if ≥5 cm.
- **Belzutifan** (HIF-2α (hypoxia-inducible factor 2α) inhibitor), FDA (Food and Drug Administration)-approved 2021 for VHL tumours.

Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes. Father and brother affected: 9-fold risk. BRCA2 gives aggressive prostate cancer. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or a history of Lynch cancer; screen with MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1 gene: WAGR (Wilms, aniridia, genitourinary anomalies, retardation) and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screening of relatives.

Diagram to draw: VHL loss leading to HIF accumulation, VEGF (vascular endothelial growth factor) production and angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

SAQ 4 Penile inversion vaginoplasty

P2 · P4 **merged from 2**

Asked as: P2 SAQ "Penile inversion vaginoplasty technique." · P4 SAQ "Penile Inversion Vaginoplasty Technique"

Introduction

- The commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.
- **Principle:** homologous penile and scrotal tissues transferred on their own blood supply.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking (fertility preservation); continue oestrogen.

- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: a vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

Tissue transfer (homologous tissues)

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial mucosal collar	Clitoral hood, inner labia minora
Dorsal bulbar urethral plate (spatulated urethra)	Vestibule
Penile shaft skin (inverted)	**Neovaginal lining**
Scrotal skin (full-thickness graft)	Extra (proximal) canal lining
Lateral scrotal skin + cord fat	Labia majora

Operative steps

`Lithotomy; tranexamic acid 1 g; catheter → harvest, defat and tubularise scrotal skin graft → bilateral orchidectomy (leave cord fat for labial bulk) → deglove penis between dartos and Buck's fascia (keep dartos with skin = blood supply) → separate dorsal neurovascular bundle + glans; excise ALL corporal erectile tissue → clitoroplasty; fix at level of adductor longus tendons → shorten and spatulate urethra; excise bulbospongiosus → canal dissection between prostate and rectum: central tendon, behind prostate, preserve Denonvilliers fascia, repeated rectal checks → invert penile skin tube + graft into canal; labiaplasty → pack canal; negative-pressure dressing`

- **Why excise all erectile tissue:** residual tissue engorges and obstructs (narrows) the canal.
- Depth depends on canal dissection, not penile length.
- **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

Postoperative care

- Pack and catheter out on about day 5.
- **Lifelong dilation**: several times daily at first, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy; continue oestrogen.

Complications and outcomes

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.
- **Rectovaginal fistula** (0.8% primary, 6.3% revision): the worst complication. Urethrovaginal fistula (0.8–3.9%).
- Urethral stricture, meatal stenosis or malposition (spraying).
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

Diagram to draw: sagittal view of the lined neovaginal canal between prostate and rectum.

Key points to remember:

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

SAQ 5 Prepucioplasty

P3 · P4 **merged from 2**

Asked as: P3 SAQ "Prepucioplasty" · P4 SAQ "Prepucioplasty."

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanoposthitis) or BXO (balanitis xerotica obliterans).

Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance.
- Treat active infection first.

Techniques

Technique	Method
Dorsal slit with transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm full retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

> **Key points to remember:**
> - Steroid first; about 1 in 6 fail.
> - Cut lengthwise, close crosswise; daily retraction after.
> - BXO: circumcision preferred.

SAQ 6 Oncogenes

P1

Asked as:

- Oncogenes — *Winter 2017 Paper I Q3f (SAQ)*
- Oncogene — *Practice Set 6 Paper I Q3f (SAQ)* (also answered in 2017-Winter Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives (driving) uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from (found in) the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Point (activating) mutation	Constitutively active protein (protein permanently "on")	RAS (bladder); FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; MET germline (germline MET) in hereditary papillary RCC (renal cell carcinoma)
Gene amplification	Increased copy number (more copies) → overexpression	MYC (prostate, bladder); HER2 (human epidermal growth factor receptor 2); AR (androgen receptor) in castration-resistant prostate cancer
Chromosomal translocation / fusion	Gene placed under a strong promoter, or fusion protein	**TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC)

Mechanism	Effect	Example
- **Also know:** promoter hypomethylation can switch oncogenes on.		

Functional classes

Class	Examples
Growth factors	PDGF (platelet-derived growth factor), sis
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma)
Signal transducers	**RAS** (binds GTP [guanosine triphosphate]; GTP-binding; mutation locks it "on"), BRAF, PI3K
Transcription factors	**MYC**, ERG, ETS family
Cell cycle / apoptosis regulators	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Normal role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function
Alleles needed	One	Usually both (**Knudson two-hit**: retinoblastoma)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; RAS (HRAS) mutually exclusive with FGFR3; HER2 in 9–12% of MIBC; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after (under) hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

Targeted therapy ("oncogene addiction")

- **Erdafitinib** (pan-FGFR inhibitor) for FGFR-altered (FGFR3/2-altered) advanced (metastatic) urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

Key points to remember:

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

SAQ 7 Premalignant conditions in urology (including premalignant lesions of penis)

P1

Asked as:

- Premalignant conditions in urology — *Summer 2021 Paper I Q3a (SAQ)*
- Premalignant lesion of penis — *Practice Set 6 Paper I Q3a (SAQ)*

Type: SAQ (10 marks)

(premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.
- Penis: a lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

Penis

Classification

1. PeIN (HPV-associated or HPV-independent)

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty) or differentiated (HPV-independent).
- **Mechanism (HPV):** viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

Clinical entity / lesion	Site, appearance / key points	Risk of invasion
Erythroplasia of Queyrat	Glans or prepuce; red velvety (glans) plaque	**Progresses in 10–33%**
Bowen disease	Shaft; scaly red (shaft) plaque	About 5% progress
Bowenoid papulosis	Shaft papules, young men; HPV 16	Does not progress; may regress

2. Other lesions

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC (squamous cell carcinoma) in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally invasive, locally destructive, rarely metastasises; may harbour SCC.

- **Also know:** cutaneous horn; PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; (chronic) phimosis; genital lichen planus; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

Evaluation

- Biopsy first: **multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

Treatment (organ-preserving)

Situation	Treatment
Prepuce (preputial lesions)	Circumcision or excision with clear margin
Glans	**Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; CO_2 too superficial); **glans resurfacing** with skin graft
Shaft	Excision; Mohs surgery
Lichen sclerosus	Clobetasol 0.05% for 3 months then maintenance; circumcision; lifelong follow-up
Buschke–Löwenstein	Wide excision; avoid radiotherapy (anaplastic change)

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

Diagram to draw: normal urothelium vs dysplasia vs CIS (umbrella cells lost); penile epithelium with full-thickness atypia above an intact basement membrane.

Key points to remember:

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

SAQ 8 Secondary malignancies in urology

P1

Asked as:

- Secondary malignancies in Urology. — *Practice Set 3 Paper I Q3f (SAQ)*
- Secondary malignancies in Uro — *Practice Set 9 Paper I Q3c (SAQ)* (also answered in Practice Set 3 Paper 1, Q3f)

Type: SAQ (10 marks)

Interpreted (read) as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder cancer**, rectal cancer, sarcoma (in-field)	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative)
Urinary diversion / augmentation	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer	RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive

Cause	Second cancer	Key facts
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	60–100% excess after cisplatin or radiotherapy, 200% (highest) after both; etoposide-related leukaemia
Paediatric tumours	In-field solid tumours, leukaemia	Wilms tumour 1% at 10 years

Mechanisms

- Radiation and alkylating (drugs) or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of (lost) immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

Diagram to draw: ureterosigmoid anastomosis showing (with) adenocarcinoma at the junction.

> **Key points to remember:**
> - Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
> - Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
> - Ureterosigmoidostomy cancer 6–29% after 10–20 years.
> - Transplant recipients: bladder cancer about 3-fold.

SAQ 9 Genetic counselling

P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3d

Assumed meaning: genetic counselling in urological practice.

Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

Urological applications

Condition	Genes	Key point
Prostate cancer	BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13	NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening
Hereditary RCC (renal cell carcinoma)	VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1	Counsel RCC at ≤46 years, bilateral or multifocal, family history
UTUC (upper tract urothelial carcinoma)	Lynch (MSH2 highest)	Screen if <60 years or Lynch-spectrum cancer history
ADPKD (autosomal dominant polycystic kidney disease)	PKD1 (chromosome 16), PKD2 (chromosome 4)	Testing selects living related donors
Cystinuria	SLC3A1, SLC7A9 (recessive)	Screen siblings
Male infertility	Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator)	AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR
DSD (differences of sex development)	e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked)	Recurrence and gonadal tumour risk

Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

Diagram to draw: three-generation pedigree with standard symbols.

Key points to remember:

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

SAQ 10 Congenital Penile curvature..

P1

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 1, Q3d

Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

Devine and Horton classification (1973)

Type	Abnormal tissue
I	Corpus spongiosum deficient (chordee without hypospadias)
II	Dysgenetic Buck's and dartos fascia; normal spongiosum
III	Dartos only
IV	Normal layers; **asymmetric corporal compliance** (true congenital curvature)
V	Congenitally short urethra (very rare; doubted)

Two clinical groups

Feature	Chordee without hypospadias (I–III)	Congenital curvature (IV)
Direction	Ventral ± torsion	Ventral, lateral (often left), rarely dorsal
Erection size	In proportion to flaccid size	**Noticeably large**
Presentation	Childhood	18–30 years, noticed at puberty
- **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening.		

Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.

- Grade: mild <30°, moderate 30–45°, severe ≥45°.

Management

- Surgery if curvature hinders intercourse or distresses. 1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum. 2. Residual curve: **midline ventral septotomy** under artificial erection. 3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe. 4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).
- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.
- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

Diagram to draw: penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

> **Key points to remember:**
> - Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
> - Congenital curvature: young man, large erection, orthotopic meatus.
> - Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

SAQ 11 Circumcision undone

P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3d

"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.

Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

Other complications

Complication	Management
Bleeding (commonest, about 0.1%), usually frenular	Pressure, cautery, silver nitrate or suture
Glanular adhesions; **skin bridges** (can cause chordee, torsion)	Office lysis or excision; steroids do not help
Excess skin removal, degloving	Heals by secondary intention with ointment; severe: grafts

Complication	Management
Trapped penis (cicatrix over glans)	Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty
Meatal stenosis (almost only after circumcision)	Meatotomy or meatoplasty
Subcoronal fistula; glans amputation; necrosis (cautery on clamp)	Hypospadias-type repair; reattach glans within 8 hours

Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

Diagram to draw: sleeve revision showing two circumferential incisions.

> **Key points to remember:**
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

SAQ 12 Noonan syndrome

P1

Type: SAQ (10 marks)
Source: Practice Set 22 SAQ Sheet, Q3a

Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".

- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

Gene	Approximate share	Note
PTPN11 (encodes SHP2 phosphatase)	About 50%	Pulmonary stenosis common
SOS1	About 10%	Milder growth and learning effects
RAF1	About 5–10%	Hypertrophic cardiomyopathy
RIT1, KRAS, others	Smaller	KRAS: more learning difficulty

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.
- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.
- Follow-up: pubertal hormones, semen analysis, fertility counselling.
- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

Diagram to draw: RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

Key points to remember:

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy.

SAQ 13 Mixed gonadal dysgenesis.

P3

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 3, Q3e (DSD management in general in 2019-Summer Paper 3, Q3f)

Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

	MGD	Ovotesticular DSD
Karyotype	45,X/46,XY	Mostly 46,XX
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Tumour risk	15–35%	About 3%

Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.

- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
- Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
- Raised female: remove or relocate the testis.
- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

Diagram to draw: asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

> **Key points to remember:**
> - 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
> - Second commonest cause of atypical genitalia after CAH.
> - Tumour risk 15–35%: remove streaks.
> - Screen for Turner features.

SAQ 14 Telomerase

P3

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 3, Q3b

Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

Role in cancer

```
Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1) -> checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier 2) -> telomerase activation -> immortal cancer cell
```

- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

Urological relevance

Organ	Finding
Bladder	TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine
Upper tract	hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage
Prostate	Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

Diagram to draw: telomere-to-cancer flowchart with two barriers.

> **Key points to remember:**
> - Telomerase = hTERT (catalytic) + hTR (RNA template).
> - About 100 base pairs lost per division; 85–90% of tumours express telomerase.
> - TERT promoter mutation is the commonest mutation in bladder cancer.
> - TRAP assay detects urinary telomerase; inflammation gives false positives.

SAQ 15 Enlist the steps in female to male gender reassignment surgery.

P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3b

Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened): - Hydrodissection; circumferential introital incision. - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**. - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either: - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation. - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

Diagram to draw: three parts of the neourethra (native, fixed, phallic).

> **Key points to remember:**
> - Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
> - No urethral lengthening without vaginectomy (fistula risk).
> - Commonest stricture: fixed–phallic anastomosis.

SAQ 16 Antioxidants in urological malignancy

P4

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 4, Q3a (prevention trials also in 2017-Summer Paper 4, Q3a)

Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

Prostate cancer

Agent	Evidence	Result
SELECT (Selenium and Vitamin E Cancer Prevention Trial)	35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo	Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)**
ATBC (Alpha-Tocopherol, Beta-Carotene)	Low-dose vitamin E in heavy smokers	Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer
Selenised yeast	Skin-cancer trial	Secondary fall in prostate cancer prompted SELECT
Lycopene (cooked tomato)	Meta-analysis: modestly lower risk, mainly advanced cancer	WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null
Green tea catechins	HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials	Large fall in progression to cancer (meta-analysis); dose impractical
Vitamin C, high-dose antioxidants	High doses act as pro-oxidants	Supplements above median → more advanced cancer

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

Kidney cancer

- No antioxidant trial has shown benefit.

Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

Diagram to draw: ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

> **Key points to remember:**
> - SELECT: no benefit; vitamin E alone increased prostate cancer.
> - ATBC benefit limited to smokers on low dose.
> - Lycopene downgraded to "no conclusion" (2018).

SAQ 17 Gene therapy in urology

P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3f (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

Vectors

Vector	Features
Adenovirus	Efficient, non-integrating, transient expression; replication-incompetent or oncolytic
Retrovirus, lentivirus	Integrate: long expression, risk of insertional mutagenesis
AAV (adeno-associated virus)	Low immunogenicity, long expression
Non-viral	Naked plasmid, liposomes, nanoparticles: safer, less efficient
Gene silencing	siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental

Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

Diagram to draw: adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

Key points to remember:

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

SAQ 18 Molecular targeted therapy in urological cancers

P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3c

Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

Target	Drugs	Key evidence
VEGF receptors	Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib	COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus
VEGF ligand	Bevacizumab	Modest PFS (progression-free survival) gain
mTOR	Temsirolimus (IV, poor risk), everolimus (oral)	Temsirolimus improved OS (overall survival) over interferon
HIF-2α	**Belzutifan**	VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

Limitations

- Resistance develops; cures rare; biomarker testing needed.

Diagram to draw: VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

Rapid Revision Sheet · Section 2

Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T) (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia (LAQ)

- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
- 17-OHP after 36 h; salt crisis around day 5–7.
- Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
- Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

46,XX male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Female circumcision (SAQ)

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

Familial / hereditary urological malignancies (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Penile inversion vaginoplasty (SAQ)

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

Prepucioplasty (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.

- BXO: circumcision preferred.

Oncogenes (SAQ)

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Premalignant conditions in urology (including premalignant lesions of penis) (SAQ)

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

Secondary malignancies in urology (SAQ)

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Genetic counselling (SAQ)

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Congenital Penile curvature.. (SAQ)

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

Circumcision undone (SAQ)

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.
- Revision needs general anaesthesia.

Noonan syndrome (SAQ)

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Mixed gonadal dysgenesis. (SAQ)

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

Telomerase (SAQ)

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Enlist the steps in female to male gender reassignment surgery. (SAQ)

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

Antioxidants in urological malignancy (SAQ)

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

Gene therapy in urology (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

Molecular targeted therapy in urological cancers (SAQ)

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 3 · Infections & Inflammation

Long answer questions (LAQ)

LAQ 1 Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder

P2 · P3 **merged from 2**

Asked as: P3 LAQ "Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management" (Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB?, Summer 2022 Paper III Q2 LAQ; Laboratory Diagnosis of Genitourinary Tuberculosis, Winter 2023 Paper III Q3f SAQ; Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm, Practice Set 6 Paper III Q3d SAQ) · P2 LAQ "Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male."

Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.

Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG (bacille Calmette–Guérin)) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; the second commonest extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**
- Untreated, it destroys the kidney and contracts the bladder.

Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by lymph and blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** macrophages, epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% it contains the infection and it stays latent; viable bacilli stay dormant.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, steroids or other immunosuppression. **Caseation** then releases bacilli into urine.

Route	Lesion
Haematogenous (principal)	Parenchymal disease: kidney, epididymis, prostate
Descending urinary	Luminal disease: ureteric stricture, cystitis, urethra
Ascending (intravesical BCG, about 1%)	Cystitis, prostatitis, epididymo-orchitis
Contiguous	From psoas, bowel

- **Renal sequence:** 1. Bacilli lodge in **periglomerular capillaries**. 2. Cortical granulomas coalesce. 3. **Caseation**; cavities open into calyces. 4. Papillary necrosis, abscess, calcification. 5. Bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.

Site	Frequency	Pathology
Kidney	80%	Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney")
Ureter	—	Mural granulomas; stricture, commonest at the **VUJ (vesicoureteric junction)**; "beaded corkscrew" or "pipestem" ureter
Bladder	—	Cystitis starts around the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts (<100 mL); **thimble bladder <20 mL**
Epididymis	Commonest genital site	Starts in globus minor; beaded vas, posterior scrotal sinus
Prostate, seminal vesicles	22–49%	Calcification, abscess; infertility

- **Key concept:** healing occurs by fibrosis, and fibrosis obstructs. Infundibular and ureteric strictures and bladder contraction can **worsen during treatment**.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.
- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest); haematuria and flank pain (about one-third); colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

Diagnosis: principles and specimen

- GUTB (genitourinary tuberculosis) is paucibacillary with intermittent shedding, so **multiple samples** are needed.
- **Hierarchy of proof:** positive culture or DNA, then caseating granulomas on histology, then clinical picture with response to treatment.
- **Specimen:** early-morning first-void urine (EMU), **3–5 samples on consecutive days**, processed at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

Laboratory tests

Test	Method	Comment / role
Urine routine	Microscopy	**Sterile (acid) pyuria** with or without microscopic haematuria in >90%
ZN (Ziehl–Neelsen) smear	Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive	Quick; sensitivity under 50%; saprophytic mycobacteria give false positives

Test	Method	Comment / role
Urine AFB (acid-fast bacilli) culture: gold standard	Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid medium: 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days; use a liquid system as well	80–90% sensitivity with ideal technique; diagnosis; gives DST (drug-susceptibility testing); **the only test that monitors treatment**
NAAT (nucleic acid amplification test): GeneXpert MTB/RIF	Automated cartridge PCR (polymerase chain reaction); about 2 hours (result in 1–2 days)	Rapid diagnosis; detects MTB and **rifampicin resistance** (MDR (multidrug-resistant) TB surrogate). **Xpert Ultra** is more sensitive. Stays positive after cure, so cannot monitor treatment
TrueNat	Indian chip-based NAAT	NTEP (National TB Elimination Programme)
LPA (line probe assay)	DNA strip hybridisation	Rifampicin, isoniazid or second-line drug resistance
LF-LAM (lateral-flow lipoarabinomannan)	Urine antigen strip, under 30 minutes; point-of-care	HIV with symptoms, CD4 <200 or advanced disease (WHO (World Health Organization) 2019)
Histopathology	Biopsy (bladder, epididymis)	Caseating granulomas; NAAT on tissue raises yield
TST (tuberculin skin test)	5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h	Proves infection only, not active disease; BCG false positive
IGRA (interferon-γ release assay)	ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT)	Unaffected by BCG; cannot separate latent and active; shows infection, not active disease

Newer diagnostic modalities

- **Liquid culture (MGIT):** about 10 days versus 4–6 weeks on LJ.
- **NAAT:** GeneXpert MTB/RIF (Xpert Ultra more sensitive), about 2 hours, detects rifampicin resistance; TrueNat; **LPA** for isoniazid resistance.
- **Urine LF-LAM:** strip, under 30 minutes; WHO 2019.
- **IGRA:** unaffected by BCG.
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

Supportive tests

- ESR, blood count, renal function, HIV, blood sugar.
- Sputum and **chest radiograph in all** (mandatory; lung TB coexists in up to one-fifth).

Imaging and endoscopy

Imaging	Findings / role
KUB (kidney–ureter–bladder) film	Calcification (24–44%); lobar "putty" kidney

Imaging	Findings / role
IVU (intravenous urography) / CT urography	**Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, ureteric strictures, calcification, small bladder; IVU replaced by CT
Ultrasound	Children, pregnancy, scrotum; follows hydronephrosis during treatment
MRU (magnetic resonance urography)	Children, pregnancy
Cystoscopy and biopsy	**"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative (yield 19–52%)
Chest radiograph	Mandatory (lung TB in up to one-fifth)

Diagnostic algorithm

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Suspicion (sterile pyuria, storage symptoms, epididymal nodule) -> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert -> chest X-ray; CT urography Positive -> treat; DST guides regimen Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue Still negative -> empirical trial if clinically compatible
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Medical management (first step in all)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with **pyridoxine** against isoniazid neuropathy. India's NTEP uses 2HRZE/4HRE.
- Complicated disease: some treat 12–24 months, or ≥6 months after surgery (P2: some extend to 9–12 months). The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor with ultrasound** for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids for strictures: anecdotal only.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

Surgical management

- About half need surgery, **after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

Problem	Procedure
Obstruction	Early **DJ (double-J) stent** (succeeds in 41%) or **PCN**
Localised polar lesion not responding	Partial nephrectomy
Non-functioning kidney + persistent disease or hypertension	**Nephrectomy** (laparoscopic preferred; ligate artery and vein separately)
PUJ (pelviureteric junction) stricture	Pyeloplasty; ureterocalicostomy
Lower ureteric stricture	Reimplantation, psoas hitch, Boari flap; ileal ureter if long
Contracted bladder	Augmentation; substitution if <20 mL

- A stricture not improving after 6 weeks of drugs needs open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

Thimble bladder in a 30-year-old man: assessment

Assess	How	Why
Disease activity	Repeat urine AFB culture; bladder biopsy	Do not reconstruct with active TB
Capacity and compliance	Cystogram, cystoscopy under anaesthesia, urodynamics	Size and type of reconstruction
Upper tracts	Ultrasound, CT urography, MCU (micturating cystourethrogram)	VUJ stricture or reflux needs reimplantation
Renal function	Creatinine, GFR (glomerular filtration rate), split function	Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min
Urethra, bladder neck, sphincter	Urethrogram, endoscopy	Orthotopic reservoir needs a patent, continent outlet
Patient	Dexterity, motivation, fertility wishes	May need CIC (clean intermittent catheterisation)

Thimble bladder: surgical options

Bladder state	Operation
Half the bladder diseased, capacity <100 mL	**Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum)
Severely contracted	Augmentation with **ileocaecum** or sigmoid
Thimble bladder (<20 mL)	**Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with poor renal function, a strictured or incompetent urethra, inability to perform CIC, active TB, or short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

Complications of bladder reconstruction

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI; **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

Follow-up

- Relapse 6.3–22% (mean about 5 years), so follow up with urine AFB culture and imaging for up to 10 years.
- After reconstruction: also creatinine, bicarbonate, B12 and ultrasound.

Diagrams to draw:

- Kidney showing caseous cavity, phantom calyx and VUJ stricture.
- Stepwise diagnostic flowchart (algorithm above) from sterile pyuria to culture and DST.
- Supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

Key points to remember:

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

LAQ 2 Interstitial cystitis / bladder pain syndrome

P2 · P4 **merged from 2**

Asked as: P4 LAQ "Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management" (Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis, Winter 2024 Paper IV Q1 LAQ; Painful bladder syndrome management, Summer 2021 Paper IV Q3d SAQ) · P2 SAQ "Interstitial cystitis (Bladder pain syndrome)."

Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).

Definitions: how they have changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (**>6 months**) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease; BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).

Phenotypes

- **Hunner lesion disease ("IC"):** about 20%, mainly over 50; true inflammation (dense lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder often histologically normal, or shows glomerulations on cystoscopy.
- Also pelvic-floor-dominant and widespread (centralised) pain phenotypes.

Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- **Associated conditions:** 75–80% have pain beyond the pelvis; IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

Recent trends in evaluation (AUA 2022), before treatment

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.

- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume (voiding) chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine with or without bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific; obsolete); routine hydrodistension (glomerulations are non-specific); routine urodynamics (optional only).
- **Biopsy** only to exclude CIS (carcinoma in situ).

ESSIC classification

Cystoscopy with hydrodistension plus biopsy, e.g. "3C":

Cystoscopy	Biopsy
X not done; 1 normal; 2 glomerulations; 3 Hunner lesion	X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis)

Differential diagnosis (exclude)

- UTI (urinary tract infection) including tuberculosis, CIS, stones, radiation or cyclophosphamide cystitis, **ketamine cystitis**, OAB, diverticulum, endometriosis.

Phenotype-based diagnosis and treatment (key trend)

Phenotype	Identified by	Treatment
Hunner lesion	Cystoscopy	Fulguration and/or triamcinolone; ciclosporin A if refractory
Anaesthetic-responsive bladder pain	Local anaesthetic challenge	Intravesical therapy
Pelvic-floor tenderness	Pelvic-floor examination	Myofascial physiotherapy
Widespread (centralised) pain	Body map, associated pain conditions	Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

Management principles (AUA 2022)

- **Old AUA (2011/2014): six tiers**, from self-care to surgery.
- **AUA 2022 dropped the tiers:** grouped options, phenotype-guided, shared decisions; multimodal care allowed early; major surgery stays last.
- Stop treatments that fail; monitor with symptom scores.

Treatment options

Group	Options
Behavioural	Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plans
Physical	**Myofascial pelvic-floor physiotherapy** if tender (Standard); **avoid Kegel exercises**
Oral	**Amitriptyline** 10 mg at night, rising to about 50 mg; hydroxyzine; cimetidine; **PPS 100 mg three times daily** (**pigmentary maculopathy**: baseline and periodic retinal checks)
Intravesical	**DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell**; heparin; alkalinised lidocaine (flares); also hyaluronic acid, chondroitin
Procedures	**HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 cmH_2O) hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation; risk of retention); SNM (sacral neuromodulation)
Refractory HL	**Oral ciclosporin A** (watch kidney function and blood pressure)
Last resort (<10%)	Cystoplasty (augmentation) or diversion with or without cystectomy; pain may persist

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral (systemic) steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS, fibromyalgia) with other specialists.

Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF (nerve growth factor)) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

Diagram to draw: cystoscopic stellate Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

> **Key points to remember:**
> - AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
> - AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
> - HL: fulguration or triamcinolone; ciclosporin if refractory.
> - Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

LAQ 3 Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management

P1

Asked as:

- Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock. — *Summer 2016 Paper I Q2 (LAQ)*
- Describe Pathophysiology of urosepsis and its management. — *Summer 2017 Paper I Q2 (LAQ)*
- Describe pzthophysiology of urosepsis and its management. — *Practice Set 11 Paper I Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q2) — *"Pzthophysiology" is read as pathophysiology.*

Type: LAQ (25 marks)

Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.
- **Urosepsis:** sepsis whose source is the urogenital tract (the urinary tract or male genital organs). It is usually Gram-negative and usually follows obstruction or instrumentation.

Pathogenesis of UTI

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: $\geq 10^5$ CFU (colony-forming units)/mL (Kass); $\geq 10^2$ in a symptomatic woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

Characteristics of common organisms

Organism	Key features
E. coli	~85% of community and ~50% of hospital UTI; commonest cause of urosepsis; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant
Proteus	Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones
Klebsiella	Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers
Pseudomonas	Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem
Enterococcus	**Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant)
S. saprophyticus	~10% of cystitis in young sexually active women
S. aureus	Haematogenous renal abscess; in urine, look for bacteraemia
Anaerobes, *Candida*	Abscesses, fistulae; *Candida* with catheters and diabetes (*Candida* in catheterised diabetics)

- **Urosepsis organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.
- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015-Summer Paper 1, Q3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

Bacteraemia, sepsis, septic shock and urosepsis

Sepsis-3 (2016)

Term (Sepsis-3, 2016)	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (of death) (prompt only)
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite (adequate) fluids; mortality >40%
- SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific.	

Causes and risk factors (urological triggers)

- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.

- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%), catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.

Pathophysiology

- **Pathophysiology (summary):** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.

1. Entry of bacteria (and endotoxin) into blood

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood (circulation) by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin (fragmenting infected stones releases endotoxin). Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney is (behaves like) an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

2. Host response `Endotoxin (LPS, lipopolysaccharide) / (or) Gram-positive wall products → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB) → cytokines TNF-α (tumour necrosis factor α), interleukins IL-1, IL-6, IL-8 → complement + coagulation activation + endothelial injury (activation)`

- **Vasodilatation:** inducible nitric oxide causes (lowers) low systemic vascular resistance ("warm" shock).
- **Capillary leak:** (endothelial) glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction (failure):** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase predisposes to (invites) secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction (failure), DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts (begins) within hours of an endourological procedure (endourology).

Investigations

- **Lactate**, blood cultures ×2 (two blood cultures), urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial (blood) gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021) Recognise (SSC: do not use qSOFA alone for screening) → Lactate → blood cultures ×2 + urine culture → IV (intravenous) broad-spectrum antibiotics within 1 hour (shock or probable sepsis; within 3 h if sepsis only possible) → 30 mL/kg (balanced) crystalloid if hypotensive or lactate ≥4 mmol/L (within 3 h if hypotensive or hypoperfused) → noradrenaline first line for MAP ≥65 (target MAP 65); add vasopressin → SOURCE CONTROL: stent or nephrostomy (equally effective), drain abscess, change catheter; stone surgery later → refractory shock (persistent vasopressor need): hydrocortisone 200 mg/day; ICU (intensive care unit) → re-measure (repeat) lactate, capillary refill → de-escalate by culture at 48–72 h

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis (equally effective in a randomised trial (Pearle, 1998)).
- Stent: suits coagulopathy; avoid high-pressure contrast injection.
- PCN: better for (suits) large or impacted stones, pyonephrosis; allows (gives) pelvic urine (for) culture.
- **No definitive stone surgery during sepsis.** Treat the stone after (once) the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (percutaneous drainage, nephrectomy if diffuse gas without fluid); prostatic abscess drainage; change or remove (the) catheter; debride Fournier gangrene.

C. Supportive care (Support)

- Oxygen and (or) ventilation, glucose control (insulin if glucose ≥180 mg/dL), transfusion (transfuse) below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

D. De-escalation and duration

- Narrow antibiotics by (to) culture at 48–72 h. Usual course (usually) 7–10 days (another source: 7–14 days); longer with an undrained focus or *S. aureus* bacteraemia; shorter courses suffice after good source control. Step down (switch) to oral therapy when (once afebrile and) stable.

Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage (delayed drainage) is the main avoidable cause of death.

Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

Diagram to draw: pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock; obstructed kidney leading to backflow, cytokine cascade and shock; obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

> **Key points to remember:**
>
> - *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
> - Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
> - Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
> - Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

LAQ 4 Describe clinical implication of AIDS on the Genito-Urinary Tract?

P1

Type: LAQ (25 marks)
Source: 2020-Summer Paper 1, Q1

Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

Stage	Features
1. Acute	2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative
2. Clinical latency	Years; usually no symptoms
3. AIDS	**CD4 <200 cells/mm³** or an AIDS-defining condition

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

Manifestations by site

Site	Manifestations
Genital skin and STDs	Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner
UTI (urinary tract infection)	About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment
Opportunistic UTI	No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus
Kidney	Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls

Site	Manifestations
GU TB	HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels
Prostate	Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals
Testis and epididymis	Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism
Voiding	LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy
Sexual function	ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders

HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

Malignancy

AIDS-defining	Non-AIDS-defining
Kaposi sarcoma (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer	Penile, anal, testicular, prostate cancers; Hodgkin lymphoma
Falls with HAART	Now about 70% of cancers in treated HIV

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.
- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.

- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.
- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

Also know: IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction; transplantation is possible with controlled HIV.

Diagram to draw: male GU tract outline labelled with the lesion at each site.

> **Key points to remember:**
> - AIDS = CD4 <200; UTI risk rises as CD4 falls.
> - HIV-TB: disseminated, less fibrosis, fewer strictures.
> - Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
> - PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

LAQ 5 Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

P3

Type: LAQ (25 marks)
Source: 2024-Winter Paper 3, Q2 (also answered in 2018-Summer Paper 1, Q1, with lymphatic anatomy)

Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

Aetiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; Culex mosquito in urban India)	Congenital: lymphangiectasia, renal lymphangioma
Rarely other parasites	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy
- Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia.	

Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).
- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

1. Confirm chyle

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial tests

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

4. Baseline: albumin, haemoglobin, renal function.

Management

`Mild -> diet + antifilarial drugs Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection` **1. Conservative**

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy; optional **omental wrap** around the hilum. 5. On-table cystoscopy confirms clear efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

4. Prevention

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
> - Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
> - Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
> - Chyluria precedes elephantiasis and is often intermittent.

Short answer questions (SAQ)

SAQ 1 Chronic (non-bacterial) prostatitis: diagnosis, classification and management

P1 · P4 **merged from 2**

Asked as: P1 SAQ "Chronic Prostatitis : diagnosis, classification and management." · P4 SAQ "Chronic non bacterial prostatitis."

Definition and classification: NIH (National Institutes of Health), 1999

Category	Description
I	Acute bacterial prostatitis
II	**Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection) with the **same organism**; well between episodes
IIIA	**CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, **inflammatory**: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen
IIIB	CP/CPPS, **non-inflammatory**
IV	Asymptomatic inflammatory prostatitis (incidental)

- Chronic non-bacterial prostatitis is now called **CP/CPPS**, NIH category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. It accounts for 90–95% of prostatitis.
- The IIIA vs IIIB split has no treatment value.

Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**.
- The prostate may not be the pain source.

Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (score 0–43).
- **DRE (digital rectal examination):** prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in about half).
- **Localisation tests:**
- **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
- **2-glass** (pre- and post-massage urine): matches the 4-glass test in >95%.
- Urinalysis and culture, PVR (post-void residual), flow rate.
- Cytology if irritative symptoms or haematuria.
- Selected patients: cystoscopy (if low flow or bladder pain), urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

Management of category II (chronic bacterial prostatitis)

- Fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)).
- Chlamydia: azithromycin. Resistant organisms: TMP-SMX (trimethoprim-sulfamethoxazole) for 3 months. Refractory: suppression or TURP (transurethral resection of prostate).

Management of category III (CP/CPPS): multimodal, UPOINT-directed

Monotherapy trials fail.

Domain	Therapy
Urinary	**α-blockers** in recent-onset (<1 year), α-blocker-naive men; antimuscarinics
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ (prostate)	**Quercetin**, **pollen extract** (small RCTs, randomised controlled trials); 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia)
Infection	Quinolone or tetracycline for ≥6 weeks only if treatment-naive and <1 year (EAU), or positive VB3; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy, myofascial trigger-point release**, biofeedback

- **Others: acupuncture** (beat sham in a large RCT: 61.5% vs 38.3% response), aerobic exercise, PTNS (percutaneous tibial nerve stimulation), shockwave therapy; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Avoid (not recommended):** TURP or radical prostatectomy for pain.

Diagram to draw: UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
> - Multimodal therapy; α-blocker plus antibiotic only early.

SAQ 2 Emphysematous pyelonephritis: prognostic factors and conservative management

P1 · P3 **merged from 2**

Asked as: P1 SAQ "EPN. Prognostic factors & conserv. mgt." · P3 SAQ "Prognostic indicators in emphysematous pyelonephritis."

Assumed meaning: EPN = emphysematous pyelonephritis.

Definition and pathogenesis

- EPN is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*.
- Most patients are **diabetic**; obstruction in about a quarter.
- **Mechanism (proposed):** high tissue glucose fermented to CO_2 in poorly perfused tissue.
- Reported mortality 19–43%.
- **CT (computed tomography)** is the imaging of choice.

Imaging-based prognostic classifications

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only (emphysematous pyelitis)
2	Renal parenchyma only, no extrarenal extension
3A	Extension into the **perinephric** space
3B	Extension into the **pararenal** space
4	**Bilateral** EPN or EPN in a **solitary kidney**

2. Wan classification (CT pattern)

Type	CT	Mortality
I	Streaky or mottled gas, **no fluid**; rapid parenchymal destruction	50–60%
II	Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system	<20%

- **Why:** fluid means a walled-off (contained) collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

Conservative management (now standard)

1. Resuscitation: fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

2. Antibiotics: broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

3. Drainage

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

How prognosis guides management (algorithm)

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction (stent/PCN) Class 1–2 -> antibiotics ± PCD (P3: antibiotics + PCD) Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds) Class 3–4, >=2 risk factors, or PCD failure -> nephrectomy (emergency or delayed)
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- Routine emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it.
- Delayed nephrectomy for a kidney that is non-functioning on renal scan.
- **Emphysematous pyelitis** (class 1, gas in the collecting system only) is often non-diabetic and often responds to antibiotics alone.

Diagram to draw: CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

> **Key points to remember:**
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
> - Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; type II <20%.
> - Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

SAQ 3 Principles of antimicrobial therapy in UTI

P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3f

Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

Killing pattern	Drugs	Dosing goal
Time-dependent (time above MIC [minimum inhibitory concentration])	β-lactams, carbapenems	Frequent doses or extended infusion
Concentration-dependent (peak:MIC)	Aminoglycosides, fluoroquinolones (FQs)	High once-daily dose

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

Empirical choices

Scenario	Drug	Duration
Cystitis (women)	Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily	5 d; single dose; 3–5 d; 3 d
Cystitis (men)	TMP-SMX or per culture	7 d (= 14 d, Drekonja 2021)
Pyelonephritis (oral)	Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily	7 d; 5 d
Pyelonephritis (intravenous)	Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily	10–14 d; oral when afebrile
Complicated / catheter UTI	Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter	7–14 d
Urosepsis	Broad-spectrum **within 1 h**, source control	–

Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

Diagram to draw: Concentration–time curve showing peak:MIC and time above MIC.

Key points to remember:

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

SAQ 4 Renal and perinephric abscess

P1

Asked as:

- Perinephric abscess — *Summer 2017 Paper I Q3c (SAQ)*
- Treatment of renal abscess — *Winter 2017 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
- **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
- **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess (paranephric beyond it).

Aetiology (perinephric abscess)

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).
- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

Clinical features

- Fever, flank pain, leucocytosis.
- Perinephric abscess is **insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

	Pyelonephritis	**Perinephric abscess**
Symptoms before admission	≤5 days	>5 days
Fever on antibiotics	Never >4 days	≥5 days (median 7)

Investigations / diagnosis (needed to plan treatment)

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.

- **CT (computed tomography): investigation of choice.** Renal abscess: round low-attenuation lesion with enhancing "ring sign". It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.
- AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

Treatment

1. Antibiotics immediately (after blood and urine cultures)

Presumed source	Empirical therapy
Haematogenous (Gram-positive)	Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*)
Ascending (Gram-negative)	IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside
Perinephric abscess	Aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture
- Switch to oral once afebrile; continue until clinical and imaging resolution.	

2. Size-based drainage (renal abscess)

Size	Management
≤3 cm	Antibiotics alone (stable patient)
3–5 cm	Antibiotics alone or drainage; no consistent pathway
>5 cm	**Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains
Any size: immunocompromised or not responding	Percutaneous drainage

- **Perinephric abscess (if present) — lower threshold:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics; drain if >3 cm. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line).
- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

4. Surgery

- Open drainage for loculated or failed (percutaneous drainage) cases; **nephrectomy** for a non-functioning, destroyed or pyonephrotic kidney, at the same time or staged.

Outcome

- Perinephric abscess: mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

Diagram to draw: coronal kidney showing renal, perinephric and paranephric spaces / abscess sites.

> **Key points to remember:**
> - Now mostly ascending Gram-negative abscess in a damaged kidney.
> - Perinephric: within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - CT is the investigation of choice; follow until resolution.
> - ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

SAQ 5 Ciprofloxacin

P1

Type: SAQ (10 marks)
Source: 2017-Summer Paper 1, Q3d

Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

> **Key points to remember:**
> - DNA gyrase and topoisomerase IV; concentration-dependent.
> - Use for pyelonephritis only if local resistance ≤10%.
> - Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

SAQ 6 Catheter Associated Urinary Tract Infections (CAUTI)

P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3c

Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.
- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥10^3 CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥10^5.

Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

Diagram to draw: catheter in bladder showing extraluminal and intraluminal routes and biofilm.

> **Key points to remember:**
> - Bacteriuria 5–10% per catheter-day; duration is the key risk.
> - Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
> - Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

SAQ 7 Complicated urinary tract infections.

P1

Type: SAQ (10 marks)
Source: 2018-Summer Paper 1, Q3f

Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

Complicating factors

Category	Examples
Obstruction	BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour
Voiding dysfunction	Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux)
Foreign body / iatrogenic	Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired
Anatomical	Diverticulum, fistula, cystocele
Host	**Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly
Other	Symptoms >7 days; recent antibiotics

- Each causes stasis, biofilm or weak defences.

Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).
- Correct the underlying cause, or infection recurs.

Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

Diagram to draw: Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

Key points to remember:

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

SAQ 8 ESBL producing enterobacterial UTI

P1

Asked as:

- ESBL producing enterobacterial UTI — *Summer 2019 Paper I Q3f (SAQ)*
- ESBL producing enterobacterial UTI. — *Practice Set 5 Paper I Q3c (SAQ)* (also answered in 2019-Summer Paper 1, Q3f)

Type: SAQ (10 marks)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel to high-prevalence areas; transrectal prostate biopsy.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone only if susceptible; single-dose aminoglycoside
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; else (otherwise) **aminoglycoside**

Infection	Options
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- **Source control:** remove or change catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; narrowest spectrum; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.
- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

Also know: AmpC (not clavulanate-inhibited) and carbapenemases **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

Diagram to draw: treatment flowchart from culture result to oral or IV (intravenous) choice.

> **Key points to remember:**
> - Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
> - Oral: nitrofurantoin, fosfomycin, pivmecillinam.
> - Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
> - Stewardship prevents spread.

SAQ 9 Lichen sclerosus (BXO, balanitis xerotica obliterans)

P1

Asked as:

- Lichen sclerosis. — *Summer 2020 Paper I Q3c (SAQ)*
- BXO — *Practice Set 10 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)
(also answered in 2020-Summer Paper 1, Q3c)

Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO). BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.

Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).

- Men: peak 30–50 years in one chapter, older men in another; also children (boys with phimosis). Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

Aetiology

- Cause unknown. Autoimmune links (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma are proposed but unproven.

Clinical features

- White, atrophic, sclerotic plaques with loss of architecture; purpura, fissures; itch, pain, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- **Mechanism of stricture:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate (lymphocytes).
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists (may coexist).
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

Management

Situation	Treatment
Skin disease	**Clobetasol propionate 0.05%** daily for 1–3 months (another source: for about 3 months), then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease
Phimosis	**Circumcision**, usually curative
Meatal stenosis	Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty
Anterior stricture	**Oral mucosal graft, never genital skin** (LS recurs in skin); staged or single stage
Panurethral or unfit	**Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation

- Recurrence 40–50% after medical treatment (recurrence is common); restricture higher than in other aetiologies (restricture rates exceed other aetiologies); treatment follows the patient's goals.

Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

Diagram to draw: penis with LS at (showing) prepuce, glans, meatus and fossa involved, proximal urethra spared.

> **Key points to remember:**
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

SAQ 10 Drug Fosfomycin

P1

Type: SAQ (10 marks)
Source: 2022-Summer Paper 1, Q3c

Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.
- Concentrations fall in renal failure; avoid at very low creatinine clearance.

Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

Uses in urology

Indication	Dose
Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America))	3 g oral sachet, single dose

Indication	Dose
Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B)	3 g single dose
MDR (multidrug-resistant) lower UTI (urinary tract infection)	Oral option avoiding IV (intravenous) therapy; repeated doses used off-label
Prophylaxis for transrectal prostate biopsy	As effective as fluoroquinolone
Recurrent UTI prophylaxis	3 g every 10 days (EAU)

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

Diagram to draw: peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

SAQ 11 Lymphogranuloma venereum

P1

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 1, Q3c

Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

Clinical stages

Stage	Timing	Features
Primary	3–30 days	Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation
Secondary (inguinal syndrome)	2–6 weeks later	Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament
Tertiary	Untreated	Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture**

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

Diagram to draw: groin with buboes above and below the inguinal ligament (groove sign).

> **Key points to remember:**
> - Serovars L1–L3; invasive and lymphotropic.
> - Proctitis is now the commonest presentation.
> - Groove sign; esthiomene; rectal stricture.
> - Doxycycline 100 mg twice daily for 21 days.

SAQ 12 Xanthogranulomatous pyelonephritis.

P1

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 1, Q3a

Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy**

macrophages (xanthoma cells).
- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous fistula**.
- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

Diagram to draw: CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

Key points to remember:

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

SAQ 13 Papillary necrosis.

P1

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 1, Q3a

Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

Causes ("POSTCARDS")

Letter	Cause
P	Pyelonephritis
O	Obstruction
S	Sickle cell disease or trait
T	Tuberculosis
C	Cirrhosis
A	Analgesic abuse
R	Renal transplant rejection; renal vein thrombosis
D	**Diabetes mellitus**
S	Systemic vasculitis
- **Also know:** infant dehydration and hypoxia; shock; renal candidiasis.	

Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.
- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

Diagram to draw: calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

SAQ 14 Cephalsporins

P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3c

Assumed meaning: cephalosporins (generations, spectrum, urological use).

Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

Generations

Generation	Examples	Spectrum	Urological use
First	Cefazolin (intravenous), cephalexin, cefadroxil (oral)	Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis*	**Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis

Generation	Examples	Spectrum	Urological use
Second	Cefuroxime, cefaclor; cefoxitin, cefotetan	Wider Gram-negative; cefoxitin and cefotetan add **anaerobes**	Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy)
Third	Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime**	Most Enterobacterales; **only ceftazidime covers *Pseudomonas***	Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis
Fourth	Cefepime	Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive	Complicated, nosocomial UTI (urinary tract infection)
Fifth	Ceftaroline	MRSA + Gram-negatives	Limited urological role

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-spectrum choice for many outpatient UTIs.

Diagram to draw: β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

> **Key points to remember:**
> - Cefazolin = standard surgical prophylaxis.
> - Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
> - No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

SAQ 15 Fungal UTI

P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3a

Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often

fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.
- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

Management

Situation	Treatment
Asymptomatic	Remove or change catheter, stop antibiotics; no antifungal
Treat asymptomatic only if	Neutropenia, very-low-birth-weight infant, before a urological procedure
Cystitis	Fluconazole 200 mg (3 mg/kg) daily for 2 weeks
Pyelonephritis	Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks
Fluconazole-resistant (*C. glabrata*, *C. krusei*)	Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine
Resistant cystitis	Amphotericin B bladder irrigation 50 mg/L
Fungus ball	PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

Diagram to draw: budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

Key points to remember:

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

SAQ 16 Antimicrobial resistance

P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3b (also answered in part in 2015-Summer Paper 1, Q3f)

Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

Types

Type	Mechanism	Relevance
Intrinsic	Species lacks the target	Predictable (enterococci and cephalosporins)
Acquired chromosomal	Mutants selected when urine level is close to MIC	Underdosing, poor compliance
Plasmid (extrachromosomal)	Horizontal transfer in bowel flora	**Commonest**; multidrug resistance after antibiotics

Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene. - **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

Diagram to draw: Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

> **Key points to remember:**
> - Plasmid-mediated resistance is commonest.
> - ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
> - TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
> - Stewardship: culture, narrow, short, no ASB treatment.

SAQ 17 Biofilm

P1

Type: SAQ (10 marks)
Source: Practice Set 18 SAQ Sheet, Q3c (also answered in part in 2015-Summer Paper 3, Q3d)

Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.

- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

Urological examples

Site	Point
Urethral catheter	Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage
Ureteric stent, nephrostomy	Stent colonised more often than urine shows; nephrostomy changed every 3 months
Penile prosthesis, AUS (artificial urinary sphincter)	Revision of an "uninfected" device: about 10% infection
Infection stones	Bacteria inside struvite; antibiotics cannot sterilise
Chronic bacterial prostatitis	Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy
Recurrent cystitis	**Intracellular bacterial communities**: biofilm-like pods inside umbrella cells

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).
- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

Diagram to draw: five-stage biofilm cycle on a catheter surface.

> **Key points to remember:**
> - Biofilm: microbes in self-made matrix on a surface.
> - Survives 1000–1500 times normal antibiotic levels.
> - Treat CAUTI by changing the catheter.
> - Revision washout cuts prosthesis infection from about 10% to 3%.

SAQ 18 Renal mucormycosis

P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3c

Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).

- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**. - **Fluconazole, voriconazole and echinocandins are inactive.** - Bilateral: antifungal plus dialysis.

Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

Diagram to draw: right-angle branching aseptate hypha invading a renal vessel.

Key points to remember:

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

SAQ 19 Hydatid disease of kidney

P1

Type: SAQ (10 marks)
Source: Practice Set 19 SAQ Sheet, Q3d

Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

WHO (World Health Organization) stage	Appearance	Activity
CE1	Unilocular, double-line wall	Active
CE2	Multivesicular, daughter cysts	Active
CE3a / CE3b	Detached membranes / daughter cysts in solid matrix	Transitional
CE4 / CE5	Solid "ball of wool" / calcified wall	Inactive

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).

- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible: 1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine). 2. Inject scolicide only if **not** communicating (chemical injury to the collecting system). 3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain. 4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

Diagram to draw: layers of a hydatid cyst with daughter cysts and a communication into a calyx.

> **Key points to remember:**
> - Kidney: third site (about 2–3%).
> - Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
> - Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

SAQ 20 Malakoplakia

P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3e

Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.

- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

	XGP (xanthogranulomatous pyelonephritis)	**Malakoplakia**
Cell	Lipid-laden foamy macrophages	Von Hansemann cells with MG bodies
Stone/obstruction	Usually (staghorn)	Absent
Laterality	Unilateral	Uni- or multifocal, bilateral
Treatment	Nephrectomy	Antibiotics; nephrectomy if unilateral, symptomatic

Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.
- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.
- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

Diagram to draw: histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

SAQ 21 Nitrofurantoin.

P4

Type: SAQ (10 marks)
Source: 2020-Summer Paper 4, Q3e

Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

Spectrum

Susceptible	Resistant
E. coli (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter*	***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter*

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

Uses and doses

Indication	Regimen
Uncomplicated cystitis (women)	**100 mg twice daily × 5 days** (modified release)
Recurrent UTI (urinary tract infection) prophylaxis, adults	50–100 mg at night, or post-coital
Children: treatment / prophylaxis	3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age)
Pregnancy	Second and third trimesters; first trimester only if no alternative; **avoid near term**

Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

Diagram to draw: spectrum table (susceptible versus resistant).

Key points to remember:

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

Rapid Revision Sheet · Section 3

Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder (LAQ)

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

Interstitial cystitis / bladder pain syndrome (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
- AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management (LAQ)

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Chronic (non-bacterial) prostatitis: diagnosis, classification and management (SAQ)

- NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
- Multimodal therapy; α-blocker plus antibiotic only early.

Emphysematous pyelonephritis: prognostic factors and conservative management (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

Principles of antimicrobial therapy in UTI (SAQ)

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

Renal and perinephric abscess (SAQ)

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- Perinephric: within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- CT is the investigation of choice; follow until resolution.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- Percutaneous drainage first line; nephrectomy if non-functioning.

Ciprofloxacin (SAQ)

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥10^3 CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

Complicated urinary tract infections. (SAQ)

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

ESBL producing enterobacterial UTI (SAQ)

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

Lichen sclerosus (BXO, balanitis xerotica obliterans) (SAQ)

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

Drug Fosfomycin (SAQ)

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

Lymphogranuloma venereum (SAQ)

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

Xanthogranulomatous pyelonephritis. (SAQ)

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

Papillary necrosis. (SAQ)

- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Cephalsporins (SAQ)

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Fungal UTI (SAQ)

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

Antimicrobial resistance (SAQ)

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

Biofilm (SAQ)

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.

- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

Renal mucormycosis (SAQ)

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

Hydatid disease of kidney (SAQ)

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Malakoplakia (SAQ)

- Defective macrophage digestion of *E. coli*.
- Michaelis–Gutmann bodies are pathognomonic.
- Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Nitrofurantoin. (SAQ)

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*